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Foundations of Morse Theory: From Sard's Theorem to Morse Inequalities

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Abstract

The main goal of this work is to provide formal and descriptive proofs of the fundamental results of Morse theory. Morse theory is a branch of differential topology that studies the topological structure of smooth manifolds by analysing smooth real valued functions defined on them, known as Morse functions. The first part of the project is devoted to the study of these functions. Specifically, we present a detailed proof of Sard's Theorem and use it to show that Morse functions are abundant and form a dense subset in the topological space of smooth functions on a manifold. In the second part, the foundational theorems of Morse theory are proved, including Morse's Lemma and results that describe the homotopy types of the level sets of a Morse function based on its non-degenerate critical points. The final part of the work establishes the Morse inequalities, which relate key topological invariants of compact manifolds–namely, the Betti numbers and the Euler characteristic–to the number of critical points of a Morse function defined on the manifold.

Resum

L'objectiu principal d'aquest treball és d'oferir demostracions formals i descriptives dels resultats fonamentals de la teoria de Morse. La teoria de Morse és una branca de la topologia diferencial que estudia l'estructura topològica de varietats diferenciables analitzant funcions suaus amb valors reals definides sobre elles, conegudes com a funcions de Morse. La primera part del projecte està dedicada a l'estudi d'aquestes funcions. En concret, presentem una demostració detallada del Teorema de Sard i el fem servir per mostrar que les funcions de Morse són abundants i formen un subconjunt dens en l'espai topològic de les funcions suaus definides sobre una varietat. En la segona part, es demostren els teoremes fonamentals de la teoria de Morse, incloent el Lema de Morse i resultats que descriuen els tipus d'homotopia dels subnivells d'una funció de Morse en funció dels seus punts crítics no degenerats. La part final del treball estableix les desigualtats de Morse, que relacionen invariants topològics claus de varietats compactes–concretament, els nombres de Betti i la característica d'Euler–amb el nombre de punts crítics d'una funció de Morse definida sobre la varietat.

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Introduction

This work is an introduction to Morse Theory, which describes the homotopy structure of smooth manifolds, mainly based on the non-degenerate critical points of smooth real valued functions defined on them, called Morse functions. Given such a function, this theory studies the structure of its underlying manifold through its level sets: subsets consisting of points that are mapped by the function to values less than or equal to a fixed real number, called the level value. One can imagine, for instance, a height function that, relative to a fixed reference direction, assigns a height to each point on a compact manifold. The corresponding level sets then consist of the points on the manifold whose height is less than or equal to a given value.

In this setup, Morse theory characterizes the homotopy types of the level sets based on the non-degenerate critical points of the function and their corresponding images, known as critical values. Specifically, consider studying the level sets of a function–such as a height function–while continuously increasing the level value. The theory shows that if no critical value is encountered during this process, then the initial and final level sets are diffeomorphic. However, if by increasing the level value we pass by a critical value, the new level set has the homotopy type of the previous one with a cell attached for each new critical point that appears. Moreover, the dimension of each attached cell is given by a numerical invariant of the corresponding critical point: its index.

The initial goal of this project was to fully understand this procedure and to comprehend why it is possible. To that end, we have developed a theoretical framework through which we have not only demonstrated this result, but also understood how Morse functions behave, and the valuable insight they offer when studying the topological structure of a given manifold. Furthermore, we have seen that Morse theory also provides understanding of the Morse functions defined on compact manifolds, by relating the number of their critical points to topological invariants through the Morse inequalities.

This project aims to give detailed and self-contained proofs of the considered results. The demonstrations are redacted to be didactic and follow logical and methodical steps. The main proofs of the text are based in books of the literature, but have been reinterpreted and expressed in a more instructive manner for those arguments that, in our view, required a more elaborate or detailed line of reasoning. Also, some of the demonstrations in this project have been carried out by the author for the sake of completeness.

Let us briefly describe the content of the work:

In the first chapter, we start by defining the basic concepts required to understand Morse theory, including Morse functions. Then, in the next section, two fundamental results of differential topology are presented: Whitney's Theorem and Sard's Theorem. We provide an exhaustive proof of Sard's Theorem, based on results from measure theory that are included and proved in the work. Although based on the reference demonstrations, we provide reinterpretations of the proofs and include representations that have been useful for understanding the demonstrations. In addition, Propositions 1.18 and 1.20, and Lemma 1.24, have been developed by the author as exercises. The following section is devoted to using these theorems to prove the abundance of Morse functions–in the sense that they can be defined with respect to almost every point of the given manifold–and that they form a dense subset in the topological space of smooth functions on a compact manifold.

In the second chapter, the main theorems of Morse theory are proven, including Morse's Lemma and the theorems which characterize the level sets of a manifold (and the whole manifold) in terms of the critical values of a Morse function. Several results required to prove these main theorems are included, along with illustrations of the sets considered in the demonstrations. The chapter ends with an illustrative example of the developed results, focusing on the classical case of a height function defined on a torus.

The final chapter of the work establishes the Morse inequalities, which relate the homotopy invariants of compact manifolds–namely, the Betti numbers and the Euler characteristic–to the number of critical points of a Morse function defined on a compact manifold. In this chapter, we formally introduce why homology vector spaces can be considered when studying a compact manifold. We conclude the work with the proof of the Morse inequalities, in their different expressions. Almost all the demonstrations included in this last chapter have been carried out by the author.

Beyond its technical aspects, this work has been a valuable learning process for me. The project has strengthened my ability to read, understand, and write proofs in differential and algebraic topology, where a geometric and visual understanding of the problem has been crucial throughout the demonstrations, while maintaining a high degree of formality and completeness.

Regarding the material reviewed, this work has helped me extend and apply the ideas I studied in previous courses on topology and differential geometry to

understand a completely new theory. I believe I have deepened my understanding of the proofs in Morse theory and gained both formal and intuitive insight into its results. Moreover, this project has also served as an introduction to algebraic topology for me, specifically homology theory, as I had no prior knowledge of the subject before undertaking this study. This exploration was conducted independently and is not included in the work itself, although I have applied the results I learned to comprehend the Morse inequalities.

Finally, I would like to mention that, although it is not included in this project, one of the objectives of this work was to understand the weak Morse inequalities from Witten's perspective, within the context of supersymmetric quantum mechanics. The motivation behind this goal was to interpret the inequalities from a different viewpoint related to physics, a degree I have studied simultaneously with my degree in mathematics. This study was carried out separately from the main work, and a session with my advisor Dr. Cardona was dedicated to discussing doubts about this alternative approach to the Morse inequalities. This content was not included in the project for three reasons: lack of time, limited space, and the absence of full mathematical rigour in the concepts developed, as it involves notions of Hodge Theory or De Rham Cohomology that I have yet to fully studied.

Mathematical Background and Notation

This project assumes familiarity with foundational topics in elementary differential geometry and algebraic topology. My prior background to this work included an undergraduate course on differential geometry. The algebraic topology concepts included in the work–essentially homology theory–are standard material in introductory texts on algebraic topology. In my case, I studied these concepts mainly from Sections 2.1 and 2.2 of *Algebraic Topology* by Allen Hatcher [4].

We now introduce the notation that will be used throughout this work to describe basic concepts from this background. From differential geometry, we use the following conventions:

(I) We denote by M a smooth manifold (which we refer to simply as a manifold). A point in M is denoted by $p \in M$, and the tangent space to M at p is denoted by T_pM. The dimension of M is denoted by $m = \dim M$.

(II) A local chart of M is denoted by (U, φ), where $U \subset M$ is an open set and $\varphi : U \to \mathbb{R}^m$ is a homeomorphism. Given $p \in M$, we write $\varphi(p) = (x_1, \ldots, x_m)$ for the local coordinates of p. A chart is said to be centered at $q \in U$ if $\varphi(q) = 0$. We denote the inverse of the chart by $\psi = \varphi^{-1} : V = \varphi(U) \to M$, which provides a local parametrization of M. The vectors

$$\left.\frac{\partial}{\partial x_1}\right|_p, \ldots, \left.\frac{\partial}{\partial x_m}\right|_p \in T_pM$$

form the coordinate basis of the tangent space at p.

(III) Given manifolds M and N, a map $F : M \to N$ is said to be smooth if it is of class C^∞. For local charts (U, φ) on M and (V, ϕ) on N, the local expression of F is given by $(\phi \circ F \circ \varphi^{-1})(x_1, \ldots, x_m)$. By abuse of notation, we may write $F(p) = F(x_1, \ldots, x_m)$.

(IV) Given $p \in M$ and $q = F(p)$, we denote by

$$d_pF : T_pM \to T_qN$$

the differential of F at p.

(V) We adopt the following notation for common structures on manifolds:

- $C^\infty(M)$ denotes the $\mathbb{R}$-algebra of smooth functions $f : M \to \mathbb{R}$.

- $\mathcal{X}(M)$ denotes the $C^\infty(M)$-module of smooth vector fields on M. A vector field $X \in \mathcal{X}(M)$ is identified with its associated derivation $X : C^\infty(M) \to C^\infty(M)$.
- $\Lambda^p(M)$ denotes the $C^\infty(M)$-module of differential p-forms on M.
- $\mathcal{I}_{(k,l)}(M)$ denotes the $C^\infty(M)$-module of (k,l)-tensor fields on M.
- A tensor field $T \in \mathcal{I}_{(k,l)}(M)$ is identified with its corresponding $C^\infty(M)$-linear map
$$T : \mathcal{X}(M)^{\otimes k} \otimes \Lambda^1(M)^{\otimes l} \to C^\infty(M),$$
or, equivalently, with the $C^\infty(M)$-multilinear map
$$T : \mathcal{X}(M)^{\times k} \times \Lambda^1(M)^{\times l} \to C^\infty(M),$$
as guaranteed by the Localization Principle.

Concerning homology theory, the following notations are used:

(I) Let A be a topological space. We denote by $H_n(A;G)$ the nth singular homology group of A with coefficients in a fixed abelian group G. When the coefficient group G is clear from the context, we simplify the notation to $H_n(A)$ instead of $H_n(A;G)$.

(II) We denote by $\tilde{H}_n(A)$ the nth singular reduced homology group of the space A.

(III) For a pair of spaces (A,B) with $B \subseteq A$, the symbol $H_n(A,B)$ denotes the nth singular relative homology group of the pair. Since
$$H_n(A,\emptyset) \cong H_n(A),$$
an absolute homology group can be viewed as a particular case of a relative homology group when $B = \emptyset$.

Chapter 1

Morse Functions

In this chapter, we introduce Morse functions, which are the central object of our study. These are real valued differentiable functions defined on smooth manifolds, which may have critical points–that is, points where the first derivative vanishes–but at which the second derivative satisfies a non-degeneracy condition. We then present Whitney's Theorem and Sard's Theorem, providing a proof of the latter based on measure theory. Both theorems, fundamental in differential topology, are used to demonstrate the abundance and density of Morse functions within the topological space of smooth functions, when the underlying smooth manifold is compact.

1.1 Definitions

In this section, we introduce some fundamental concepts of Morse theory. The study is based on the critical points and values of smooth functions, which we now define.

Definition 1.1. Let M and N be manifolds and $F : M \to N$ be a smooth function. We say that $q \in F(M)$ is a **critical value** for F if exists $p \in M$ such that $F(p) = q$ and F is neither an immersion nor a submersion at p, i.e, d_pF is neither injective nor exhaustive. We say that such p is a **critical point** for F.
If $q \in N$ is not a critical value, then we say that it is a **regular value** for F.

Remark 1.2. (I) The condition d_pF *is neither injective nor exhaustive* is equivalent to saying that the differential does not have maximal rank.

(II) Note that regular values are not necessarily in the image of F and that non-critical points can have critical values as image.

(III) For $f \in C^\infty(M)$, the condition for $a \in \mathbb{R}$ to be a critical value can be simply expressed as: There exists $p \in M$ such that $f(p) = a$ and $d_pf \equiv 0$.

For a local coordinate system $(x_1, \ldots, x_m)$, this can be expressed as $\nabla f(p) \equiv 0$, where

$$\nabla f(p) = \left(\frac{\partial f}{\partial x_1}(p), \ldots, \frac{\partial f}{\partial x_m}(p) \right). \tag{1.1}$$

The following subset is fundamental in the context of Morse theory:

Definition 1.3. : Let M be a smooth manifold, $f \in C^\infty(M)$ and $a \in \mathbb{R}$. We define the **level set** of f with a **level value** a as:

$$M^a := f^{-1}(-\infty, a] = \{p \in M : f(p) \leq a\}.$$

Remark 1.4. If a is not a critical value of f, then it follows from the Implicit Function Theorem that M^a is a smooth manifold with boundary. The boundary is $f^{-1}(a)$, which, if non-empty, is a smooth submanifold of M by the Regular Value Theorem.

To understand the behaviour of a smooth function at a critical point, it is essential to introduce the following concept.

Definition 1.5. : Let M be a smooth manifold, $f \in C^\infty(M)$ and $p \in M$ be a critical point of f. Let $v, w \in T_pM$ and $\overline{v}, \overline{w} \in \mathcal{X}(M)$ such that $\overline{v}_p = v$ and $\overline{w}_p = w$.
We define the **Hessian** of f at p as the symmetric bilinear functional:

$$\begin{aligned} H_p : T_pM \times T_pM &\to \mathbb{R} \\ (v, w) &\to \overline{v}_p(\overline{w}(f)) \end{aligned} \tag{1.2}$$

Proposition 1.6. The Hessian is well defined as a symmetric bilinear functional.

Proof. a) Symmetric: Note that $H_p(v, w) - H_p(w, v) = \overline{v}_p(\overline{w}(f)) - \overline{w}_p(\overline{v}(f)) = [\overline{v}, \overline{w}]_p(f)$ where $[\overline{v}, \overline{w}] \in \mathcal{X}(M)$ is the Lie bracket of $\overline{v}$ and $\overline{w}$. For any coordinate system $(x_1, \ldots, x_m)$ in a neighbourhood of p, we can express $[\overline{v}, \overline{w}]_p = \sum_{i=1}^m a^i(p) \frac{\partial}{\partial x_i}\Big|_p$ where $a^i \in C^\infty(M)$ for every i. Since p is a critical point of f, we have $\frac{\partial f}{\partial x_i}\Big|_p = 0$ for every i, therefore $[\overline{v}, \overline{w}]_p(f) = 0$.

b) Well-defined: Since $\overline{v}_p(\overline{w}(f)) = v(\overline{w}(f))$ is independent of the extension $\overline{v}$ and $\overline{w}_p(\overline{v}(f)) = w(\overline{v}(f))$ is independent of $\overline{w}$, symmetry implies that H_p is well defined.

c) Bilinear: For any coordinate system $(x_1, \ldots, x_m)$ in a neighbourhood of p, let us express $v = \sum_{i=1}^m v^i \frac{\partial}{\partial x_i}\Big|_p$ and $w = \sum_{j=1}^m w^j \frac{\partial}{\partial x_j}\Big|_p$ where $v^i, w^j \in \mathbb{R}$ for every i and j. We can take the extension $\overline{w} = \sum_{j=1}^m w^j \frac{\partial}{\partial x_j}$ where $w^j \in C^\infty(M)$ denotes the constant function. Then:

$$H_p(v,w) = \overline{v}_p(\overline{w}(f)) = v(\sum_{j=1}^m w^j \frac{\partial f}{\partial x_j}) = \sum_{i=1}^m \sum_{j=1}^m v^i w^j \frac{\partial^2 f}{\partial x_i \partial x_j}(p).$$

□

Remark 1.7. (I) Observe from c) that the square matrix $\left(\frac{\partial^2 f}{\partial x_i \partial x_j}(p)\right)$ represents the Hessian with respect to the basis $\{\frac{\partial}{\partial x_1}\Big|_p, \ldots, \frac{\partial}{\partial x_m}\Big|_p\}$ of T_pM.

(II) Note how in the proof, we use that the point where we define the Hessian is a critical point. The Hessian is not defined at a general point of the manifold. The second-order derivative at a general point can be computed, but it depends on the chart, which is not the case when the point is critical.

The Hessian allows us to distinguish between different types of critical points. To do so, we introduce the following concepts:

Definition 1.8. : Let H be a bilinear functional on a vector space E.
We define the **index** of H as the maximal dimension of a subspace of E on which H is negative definite.
We define the **nullity** of H as the dimension of the null-space, that is, $\mathrm{Ker}(H) = \{v \in E : H(v,w) = 0 \ \forall w \in E\}$.

Definition 1.9. : Let M be a smooth manifold, $f \in C^\infty(M)$, $p \in M$ be a critical point of f and H_p be the Hessian of f at p.
We define the **index of f at p** as the index of H_p.
We say p is a **non-degenerate** critical point of f if H_p has nullity equal to 0. Critical points of f that do not satisfy this condition are called **degenerate** critical points of f.

Remark 1.10. For some local coordinate system $(x_1, \ldots, x_m)$ in a neighbourhood of p, the non-degeneracy condition is equivalent to the non-singularity of the Hessian matrix in the given coordinate system, $\left(\frac{\partial^2 f}{\partial x_i \partial x_j}(p)\right)$.

Smooth functions that do not have degenerate critical points are called Morse functions.

Definition 1.11. : Let M be a smooth manifold. We say $f \in C^\infty(M)$ is a **Morse function** if all its critical points are non-degenerate.

1.2 Sard's Theorem and Whitney's Theorem

In this section, we introduce two fundamental theorems of differential geometry: Whitney's Theorem and Sard's Theorem. These theorems will be useful for further proofs concerning Morse functions. Additionally, Sard's Theorem characterizes the critical values of smooth functions, which are particularly relevant in Morse theory.

We first state Whitney's Theorem, which embeds compact manifolds into the Euclidean space. A proof of this result can be found in standard textbooks on differential topology, and was presented during the differential geometry course given by Dr. Ricardo García López.

Theorem 1.12. (Whitney's Embedding Theorem) Let M be a compact manifold. There exists $n \in \mathbb{N}$ such that M is a differential submanifold of the Euclidean space $\mathbb{R}^n$.

The result also holds for arbitrary smooth manifolds, and further theorems provide minimal possible values of n. In particular, the strong Whitney Embedding Theorem gives $n = 2m$ as the optimal minimum.

Next, we introduce Sard's Theorem, which asserts that the set of critical values of a smooth function has measure zero. Before stating the theorem, we first define what is meant by a set of measure zero.

Definition 1.13. Let $a = (a_1, \dots, a_n)$, $b = (b_1, \dots, b_n) \in \mathbb{R}^n$ such that $a_i < b_i$ for $1 \leq i \leq n$. We define a **rectangular solid** of $\mathbb{R}^n$ as $S = S(a, b) = (a_1, b_1) \times \cdots \times (a_n, b_n) \subseteq \mathbb{R}^n$.
We define the **volume** of the rectangular solid as $V(S) = (b_1 - a_1) \dots (b_n - a_n) \in \mathbb{R}^+$.
The rectangular solid S is said to be a **cube** if $b_i - a_i$ has the same value for $1 \leq i \leq n$.

Definition 1.14. A subset $A \subseteq \mathbb{R}^n$ has **measure zero** if for all $\varepsilon > 0$, there exists a countable collection of rectangular solids $\{S_i\}_{i \in \mathbb{N}}$ which cover A, i.e, $A \subseteq \bigcup_i S_i$, and such that $\sum_i V(S_i) < \varepsilon$.

The notion of measure zero can be naturally extended to subsets of a manifold.

Definition 1.15. Let M be a smooth manifold and C a subset of M. We say that C has **measure zero in the manifold** M if for every local parametrization (U, ψ) of M, $\psi^{-1}(C) \subseteq U$ has measure zero in $\mathbb{R}^m$.

After clarifying the notion of measure zero, we are prepared to state Sard's Theorem.

Theorem 1.16. (Sard's Theorem) Let M and N be manifolds and $F : M \to N$ be a smooth function. The set of critical values of F has measure zero in N.

This result can be restated in terms of the regular values as follows: almost every point in N is a regular value of F; that is, all points in N are regular values except for a set of measure zero. Moreover, we have the following corollary:

Corollary 1.17. The regular values of any smooth function $F : M \to N$ are dense in N. In fact, for any countable collection of smooth maps $\{F_i : M_i \to N\}_{i \in \mathbb{N}}$, the points of N that are simultaneously regular values for all of the F_i are dense in N.

The first statement holds because a set of measure zero cannot contain a non-empty open set, and the second holds because a countable union of sets of measure zero also has measure zero. These properties of measure zero sets will be proven later.

For the remainder of the section, we will focus on the proof of Sard's Theorem. We will provide a proof based on the appendix of *Differential Geometry* by Victor Guillemin and Alan Pollack [3], with personal annotations, reinterpretations, and additional proofs not included in the original work.

The first part of the proof relies on measure theory. We start by introducing some results.

Proposition 1.18. The rectangular solids S_i in Definition 1.14 may be taken to be cubes.

Proof. Let $S_i = (a_{i1}, b_{i1}) \times \cdots \times (a_{in}, b_{in})$ be a rectangular solid of a measure zero covering of A. As $\mathbb{Q}$ is dense in $\mathbb{R}$, we can always take $a_{ij}, b_{ij} \in \mathbb{Q}$ for all i, j without compromising the validity of the measure zero condition.
Let i be fixed and $j = 1, \ldots, n$. Let $b_{ij} - a_{ij} = \frac{p_{ij}}{q_{ij}}$ where $p_{ij}, q_{ij} \in \mathbb{N}$. Let us partition the interval (a_{ij}, b_{ij}) in $m_{ij} := p_{ij} \prod_{k=1 k \neq j}^{n} q_{ik} = \frac{p_{ij}}{q_{ij}} \prod_{k=1}^{n} q_{ik}$ intervals of length $l_i := \frac{1}{\prod_{k=1}^{n} q_{ik}}$, which does not depend on j.

Let us express the partition of the interval as: $(a_{ij}, b_{ij}) = \bigcup_{k=1}^{m_{ij}-1} (a_{ij} + (k-1)l_i, a_{ij} + kl_i] \cup (b_{ij} - l_i, b_{ij})$ which we can write as: $(a_{ij}, b_{ij}) = \bigcup_{k=1}^{m_{ij}} A_{ijk}$ where $A_{ijk} \subseteq (a_{ij}, b_{ij})$ such that $A_{ijk} \cap A_{ijl} = \emptyset$ for all $k \neq l$ and $V(A_{ijk}) = l_i$ for all j, k. Note that the

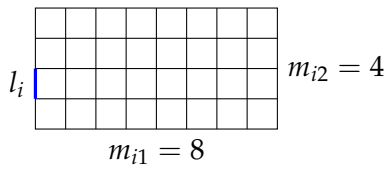

Figure 1.1: Representation of the subdivision of the rectangle S_i into squares in the case $n = 2$.

intervals A_{ijk} are not of the same type, but this does not have implications in terms of their lengths, since $V((a,b)) = V([a,b)) = V((a,b]) = V([a,b]) = b - a$ for all $a < b$.

We can express the rectangular solid S_i as the union of cubes of the form $A_{i1k_1} \times \cdots \times A_{ink_n}$ for $1 \leq k_j \leq m_{ij}$. The number of cubes is $\tilde{m}_i := m_{i1} \dots m_{in}$ all with the same volume $(l_i)^n$. Let us denote these cubes as $C_{i1}, \dots, C_{i\tilde{m}_i}$ which satisfy $C_{ij} \cap C_{il} = \emptyset$ for all $j \neq l$ and cover S_i with the same volume: $S_i = \bigcup_{j=1}^{\tilde{m}_i} C_{ij}$ and $V(S_i) = \sum_{j=1}^{\tilde{m}_i} V(C_{ij})$. Therefore, $\{C_{i1}, \dots, C_{i\tilde{m}_i}\}_{i \in \mathbb{N}}$ is a countable measure zero covering of A by cubes. □

Proposition 1.19. Subsets of sets with measure zero have measure zero, and countable unions of sets with measure zero have measure zero.

Proof. The first statement follows trivially from the definition. To prove the second statement, let $\{C_i\}_{i \in \mathbb{N}}$ be a countable collection of sets with measure zero. Given $\varepsilon > 0$, for each $i \in \mathbb{N}$ let us consider a covering $\{S_j^i\}_{j \in \mathbb{N}}$ of C_i, such that $\sum_{j=1}^{\infty} V(S_j^i) < \frac{\varepsilon}{2^i}$. Then, the countable collection $\{S_j^i\}_{i,j \in \mathbb{N}}$ covers $\bigcup_i C_i$ and has total volume $\sum_{i,j} V(S_j^i) < \varepsilon \sum_{i=1}^{\infty} \frac{1}{2^i} = \varepsilon$. □

The existence of sets of measure zero is straightforward to demonstrate, as shown in the following proposition:

Proposition 1.20. Let us denote the natural copy of $\mathbb{R}^k$ inside $\mathbb{R}^n$, for $k < n$, by $\tilde{\mathbb{R}}^k := \{(x_1, \dots, x_k, 0, \dots, 0) \in \mathbb{R}^n\}$. Then, $\tilde{\mathbb{R}}^k$ has measure zero in $\mathbb{R}^n$ for every $k < n$.

Proof. We will prove that for every $n \in \mathbb{N}$, $\tilde{\mathbb{R}}^{n-1}$ has measure zero in $\mathbb{R}^n$. Since subsets of sets of measure zero have measure zero (by Proposition 1.19), this will prove the proposition.
For every closed set A of $\mathbb{R}^n$, A can be expressed as a countable union of compacts. Indeed, since Euclidean space is σ-compact, that is, $\mathbb{R}^n$ can be expressed as a countable union of compacts $\{C_i\}_{i \in \mathbb{N}}$ (for example, the discs $\{\overline{B(0,i)}\}_{i \in \mathbb{N}}$), then A

can be expressed as a countable union of sets $\{C_i \cap A\}_{i\in\mathbb{N}}$, which are closed and contained in compact sets, and therefore compact.
Since $\tilde{\mathbb{R}}^{n-1} = \{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_n = 0\}$ is a closed set, by Proposition 1.19 is suffices to prove that every compact subset of $\tilde{\mathbb{R}}^{n-1}$ has measure zero. Let C be a compact subset of $\mathbb{R}^{n-1}$. Since C is bounded, there exists a rectangular solid S of $\mathbb{R}^{n-1}$ such that $C \subseteq S$. Let $\varepsilon > 0$ and consider the rectangular solid of $\mathbb{R}^n$, $S' = S \times (-\frac{\varepsilon}{2V(S)}, \frac{\varepsilon}{2V(S)})$, which contains $C \times \{0\}$ (an arbitrary compact subset of $\tilde{\mathbb{R}}^{n-1}$). Then, S' is a measure zero covering of $C \times \{0\}$, since $V(S') = V(S)\frac{\varepsilon}{V(S)} = \varepsilon$.

□

Evidently, sets that do not have measure zero also exist. For instance, a rectangular solid does not have measure zero, which in particular implies that a set of measure zero cannot contain a non-empty open set, since any non-empty open set contains a rectangular solid. To prove this condition, it suffices to show that every covering of a solid has volume greater than the finite volume of the solid. We now give a formal proof of this statement:

Proposition 1.21. Let S be a rectangular solid in $\mathbb{R}^n$, and $\{S_j\}_{j\in\mathbb{N}}$ a covering of the closure $\overline{S}$ by rectangular solids. Then, $\sum_j V(S_j) \geq V(S)$.

Proof. Let $(z_1, \ldots, z_n) \in \mathbb{N}^n$ be an integer point of $\mathbb{R}^n$. Let N_A be the number of integer points in a subset A of $\mathbb{R}^n$. Let $S = S(a, b)$ where $a = (a_1, \ldots, a_n)$, $b = (b_1, \ldots, b_n) \in \mathbb{R}^n$ and $S_j = S_j(a^j, b^j)$ where $a^j = (a_1^j, \ldots, a_n^j)$, $b^j = (b_1^j, \ldots, b_n^j) \in \mathbb{R}^n$ for every $j \in \mathbb{N}$.
Note that, for every i, $b_i - a_i - 1 \leq N_{(a_i,b_i)} < b_i - a_i + 1$, and let us assume, for the moment, that the lengths of the sides of S are greater than one, $b_i - a_i > 1$ (such that $N_{(a_i,b_i)} \geq 1$). Then, $\prod_{i=1}^n (b_i - a_i - 1) \leq N_S < \prod_{i=1}^n (b_i - a_i + 1)$.
Since $\overline{S}$ is compact, we may assume $\{S_j\}_{j=1}^N$ is a finite covering of $\overline{S}$. Then, $N_S \leq \sum_{j=1}^N N_{S_j}$, and thus $\prod_{i=1}^n (b_i - a_i - 1) \leq \sum_{j=1}^N \prod_{i=1}^n (b_i^j - a_i^j + 1)$.

For every $\lambda \in \mathbb{R}^+$, let $\lambda S(a, b) := S(\lambda a, \lambda b)$, which satisfies $V(\lambda S(a, b)) = \lambda^n V(S(a, b))$. Note that $\overline{\lambda S}$ is covered by $\{\lambda S_j\}_{j=1}^N$. If we now assume that the sides of S are of arbitrary length, there always exists a sufficiently large λ such that all the sides of λS are of length greater than one. Then, the last inequality holds for λS: $\prod_{i=1}^n (\lambda b_i - \lambda a_i - 1) \leq \sum_{j=1}^N \prod_{i=1}^n (\lambda b_i^j - \lambda a_i^j + 1)$. By dividing the inequality by λ^n and making $\lambda \to \infty$ we get the inequality of volumes we were looking for.

□

Let us now prove the measure zero form of Fubini's Theorem, which we will use in the proof of Sard's Theorem. First, we introduce the following concept.

Definition 1.22. Let $n, k, l \in \mathbb{N}$ such that $n = k + l$ so we write $\mathbb{R}^n = \mathbb{R}^k \times \mathbb{R}^l$. For each $c \in \mathbb{R}^k$, we define the **vertical slice** as the set $V_c := \{c\} \times \mathbb{R}^l$.
We say a subset $\{c\} \times A \subseteq V_c$ has measure zero in V_c if A has measure zero in R^l.

Theorem 1.23. (Fubini's Theorem for measure zero) Let A be a closed subset of $\mathbb{R}^n$ such that $A \cap V_c$ has measure zero in V_c for all $c \in \mathbb{R}^k$. Then A has measure zero in $\mathbb{R}^n$.

Since A is closed in the Euclidean space $\mathbb{R}^n$, it can be expressed as a countable union of compacts (by σ-compactness). Then, by the Proposition 1.19, we may assume A is compact. Furthermore, by induction on k, it is enough to prove the case $k = 1$, $l = n - 1$ (which is in fact the case we need to prove Sard's Theorem). Before showing Fubini's Theorem, we present the following two lemmas:

Lemma 1.24. Let $\{S_i\}_{i=1}^N$ be a covering of the closed interval $[a, b]$ in $\mathbb{R}$. Then there exists another covering $\{S'_j\}_{j=1}^M$ such that each S'_j is contained in some S_i and $\sum_{j=1}^M V(S'_j) < 2(b - a)$.

Proof. Let $\{S_i\}_{i=1}^N = \{(a_i, b_i)\}_{i=1}^N$ be a covering of $[a, b]$ ($[a, b] \subseteq \bigcup_{i=1}^N (a_i, b_i)$ and $(a_i, b_i) \cap [a, b] \neq \emptyset$ for $1 \leq i \leq N$).
Let $b_{i_1} := \max\limits_{1 \leq i \leq N} \{b_i : a_i < a\}$ which satisfies $b_{i_1} > a$ as (a_{i_1}, b_{i_1}) must intersect $[a, b]$.
Suppose $b_{i_1} \leq b$. If not, put $M = 1$.
Let $b_{i_2} := \max\limits_{1 \leq i \leq N} \{b_i : a \leq a_i < b_{i_1}\}$. Note that $b_{i_2} > b_{i_1}$ because if $b_{i_2} \leq b_{i_1}$ then $b_i \leq b_{i_1}$ for $a \leq a_i < b_{i_1}$, which implies $\tilde{a} := \min\limits_{1 \leq i \leq N} \{a_i : b_i > b_{i_1}\}$ is greater or equal than b_{i_1}, contradicting the fact that $\{(a_i, b_i)\}_{i=1}^N$ covers $[a, b]$, since b_{i_1} (at least) will not be covered. Suppose $b_{i_2} \leq b$. If not, put $M = 2$.

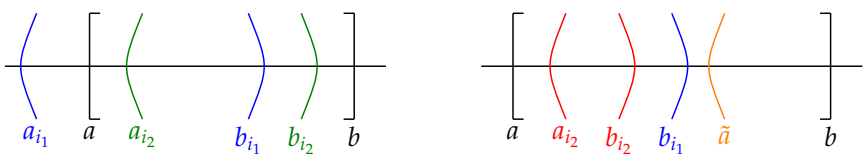

Figure 1.2: Schematic representation of the nested intervals. The left figure corresponds to the actual nesting, while the right figure shows the contradictory configuration, where the region between b_{i_1} and $\tilde{a}$ is the uncovered region.

We can repeat this same argument, defining $b_{i_k} := \max\limits_{1 \leq i \leq N} \{b_i : a_{i_{k-1}} \leq a_i < b_{i_{k-1}}\}$ which is greater than $b_{i_{k-1}}$ for $k \geq 3$. Since $\{(a_i, b_i)\}_{i=1}^N$ is a finite covering of $[a, b]$, eventually we will reach a $M \leq N$ such that $b_{i_M} > b$.
Now, note that by construction: $\sum_{j=1}^{M-1} V((a_{i_j}, b_{i_j}) \cap (a_{i_{j+1}}, b_{i_{j+1}}) < b - a$ where all

the intersections are contained in $[a,b]$. Then, we have $\sum_{j=2}^{M-1} V((a_{i_j}, b_{i_j})) + (b_{i_1} - a) + (b - a_{i_M}) = c[2(b-a)]$ where $c < 1$. Let us choose a $\delta > 0$ such that $\delta < 2(1-c)(b-a)$ and the following inclusions hold: $S_1' := (a - \frac{\delta}{2}, b_{i_1}) \subseteq S_{i_1}$ and $S_M' := (a_{i_M}, b + \frac{\delta}{2}) \subseteq S_{i_M}$. Let $S_j' := S_{i_j}$ for $2 \leq j \leq M-1$. Then, $\{S_j'\}_{j=1}^M$ is a covering of $[a,b]$ by construction, which satisfies $\sum_{j=1}^M V(S_j') = 2c(b-a) + \delta < 2(b-a)$.

□

Lemma 1.25. Let A be a compact subset of $\mathbb{R}^n$ and $c \in \mathbb{R}$. Suppose that $A \cap V_c$ is contained in an open set $\{c\} \times U$ of V_c. Then for any suitably small (closed) interval I about c in $\mathbb{R}$, $A \cap V_I$ is contained in $I \times U$, where $V_I := I \times \mathbb{R}^{n-1}$.

Proof. Let us suppose $A \cap V_c \subseteq \{c\} \times U$ where U is an open set of $\mathbb{R}^{n-1}$, but there exists a suitably small interval I about c such that $A \cap V_I$ is not contained in $I \times U$. Then, there exists a sequence $(c_j, x_j) \in A \cap V_I$ such that $c_j \to c$ but $x_j \notin U$ for all $j \in \mathbb{N}$. Since A is compact, V_I is closed and $A \cap V_I \subseteq A$ is closed, then $A \cap V_I$ is compact. By sequential compactness, there exists a convergent subsequence $(c_{j_k}, x_{j_k}) \in A \cap V_I$ with limit $(c, x) \in A \cap V_c \subseteq \{c\} \times U$ which implies that there exists $k_0 \in \mathbb{N}$ such that $x_{j_k} \in U$ for $k \geq k_0$ which is a contradiction. □

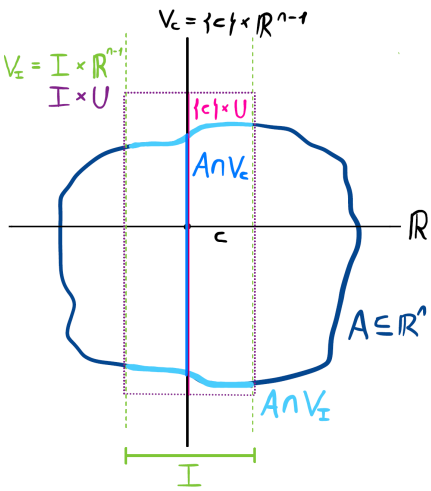

Figure 1.3: Illustration of the sets described in the Lemma 1.25. For A and its intersections, we represent only their boundaries.

We can now prove Fubini's Theorem:

Proof of Theorem 1.23. Since A is compact, we may choose an interval $I = [a,b]$ such that $A \subseteq V_I$. Let us fix $c \in I$. By hypothesis, $A \cap V_c$ has measure zero in V_c. Since A is compact, V_c is closed and $A \cap V_c \subseteq A$ is closed, then $A \cap V_c$

is compact. Then, there exists a finite collection $\{S_i(c)\}_{i=1}^{N_c}$ of rectangular solids of dimension $n-1$ such that they cover $A \cap V_c$: $A \cap V_c \subseteq \bigcup_{i=1}^{N_c}\{c\} \times S_i(c)$ and $\sum_{i=1}^{N_c} V(\{c\} \times S_i(c)) < \frac{\varepsilon}{2(b-a)}$ for all $\varepsilon > 0$.
Note that the union $\bigcup_{i=1}^{N_c}\{c\} \times S_i(c)$ is an open of V_c. Then, by applying Lemma 1.25, we can take an interval $J(c)$ about c in I such that $A \cap V_{J(c)} \subseteq \bigcup_{i=1}^{N_c} J(c) \times S_i(c)$. Therefore, $\{J(c) \times S_i(c)\}_{i=1}^{N_c}$ is a covering of $A \cap V_{J(c)}$.
We can consider the covering $\{\text{int}(J(c))\}_{c\in I}$ of the compact I, which has a subcovering $\{\text{int}(J(c_k))\}_{k=1}^{N}$ for some $c_k \in I$. Thus, $\{J(c_k)\}_{k=1}^{N}$ is a covering of I by (closed) intervals. Then, by Lemma 1.24 there exists another covering of I, $\{J'_j\}_{j=1}^{M}$, with $\sum_{j=1}^{M} V(J'_j) < 2(b-a)$ and $J'_j \subseteq J(c_{k_j})$ for some $1 \leq j \leq M$ and $1 \leq k_j \leq N$.
Since $A \subseteq V_I$, we have that $\{J'_j \times S_i(c_{k_j})\}_{1\leq j\leq M, 1\leq i\leq N_j}$, where $N_j = N_{c_{k_j}}$, is a covering of A by rectangular solids of dimension n, which satisfies:

$$\sum_{j=1}^{M}\sum_{i=1}^{N_j} V(J'_j \times S_i(c_{k_j})) < 2(b-a)\frac{\varepsilon}{2(b-a)} = \varepsilon.$$

□

In Sard's Theorem, we discuss measure zero in a manifold. To work with this concept, we need the following propositions from measure theory:

Proposition 1.26. Let U be an open set of $\mathbb{R}^n$, and let $f : U \to \mathbb{R}^n$ be a smooth function. If $A \subseteq U$ is of measure zero, then $f(A)$ is of measure zero.

Proof. Let A be of measure zero such that $\overline{A}$ is contained in U. Since $\overline{A}$ is closed, it can be expressed as a countable union of compacts, each of them with measure zero (by σ-compactness and Propostion 1.19). If we prove that the image of these compacts by f has measure zero, then $f(A)$ also has measure zero (by Proposition 1.19). Hence, we assume that $\overline{A}$ is compact.
Let W be an open neighbourhood of A such that $\overline{W}$ is compact and contained in U. Since f is smooth in $\overline{W}$, we can apply the Mean Value Theorem together with the Weierstrass Theorem to assume the existence of a constant $M > 0$ such that $\|f(x) - f(y)\| < M\|x - y\|$ for all $x, y \in \overline{W}$. Let $S \subseteq W$ be a cube of side $\delta > 0$. Then, for all $x, y \in S$, we have: $\|x - y\| = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2} < \delta\sqrt{n}$ and therefore: $\|f(x) - f(y)\| < M\delta\sqrt{n}$. By fixing some $x_0 \in S$, we have that $f(S) \subseteq B(f(x_0), M\delta\sqrt{n})$ and then $f(S)$ is contained in a cube S' with volume $V(S') < (2M\delta\sqrt{n})^n = M'V(S)$ with $M' = (2M\sqrt{n})^n > 0$ a constant (valid for every possible cube, since it does not depend on δ).
By Proposition 1.18 we can take a (finite) measure zero covering of A made by cubes $\{S_i\}_{i=1}^{N}$ such that $\sum_{i=1}^{N} V(S_i) < \frac{\varepsilon}{M'}$ for all $\varepsilon > 0$. For $1 \leq i \leq N$, we have that

$f(S_i) \subseteq S_i'$ with S_i' a cube with volume $V(S_i') < M'V(S_i)$.
Therefore, $\{S_i'\}_{i=1}^N$ is a covering of $f(A)$ with $\sum_{i=1}^N V(S_i') < M' \sum_{i=1}^N V(S_i) < \varepsilon$.

□

Proposition 1.27. A sufficient condition for a subset C of a manifold M to have measure zero is that it can be covered by the images of some collection of local parametrizations ψ_α such that $\psi_\alpha^{-1}(C)$ has measure zero for each α.

Proof. Let $\psi : U \to M$ be a local parametrization of M such that $\psi^{-1}(C) \subseteq U$ non-empty has measure zero. Then, if $\phi : V \to M$ is another local parametrization, $\phi^{-1}(C \cap \psi(U) \cap \phi(V))$ also has measure zero.
Indeed, let us assume a non-trivial case $\phi^{-1}(C \cap \psi(U) \cap \phi(V)) \neq \emptyset$. Then, one can consider the mapping:

$$\psi^{-1}(C \cap \psi(U) \cap \phi(V)) \xrightarrow{\psi} C \cap \psi(U) \cap \phi(V) \xrightarrow{\phi^{-1}} \phi^{-1}(C \cap \psi(U) \cap \phi(V)).$$

Since $\psi^{-1}(C \cap \psi(U) \cap \phi(V))$ has measure zero (since it is contained in $\psi^{-1}(C)$) and $\phi^{-1} \circ \psi$ is differentiable, by Proposition 1.26, $\phi^{-1}(C \cap \psi(U) \cap \phi(V))$ also has measure zero. Then, by covering C with the images of some parametrizations ψ_α such that $\psi_\alpha^{-1}(C)$ has measure zero for each α, every other family of parametrizations covering C satisfies the same condition. This is sufficient to prove that C has measure zero (according to Definition 1.15). □

Remark 1.28. Note that this condition is weaker than the definition. Equivalently, can be stated by saying that for every point in C there exists a local parametrization ψ for which $\psi^{-1}(C)$ has measure zero.

We are now ready to prove Sard's Theorem:

Theorem 1.29. (Sard's Theorem) Let M and N be manifolds, $F : M \to N$ be a smooth function, and let C be the set of critical points of F in M. Then $F(C)$ has measure zero in N.

Proof. Since M and N are in particular topological manifolds, they satisfy the Second Axiom of Countability, that is, they have a countable basis of open sets. Also, since they are smooth manifolds, every point has an open neighbourhood which is diffeomorphic to the Euclidean space. Therefore, we can find a countable collection of charts $\{((U_i, \psi_i), (V_i, \phi_i))\}_{i \in \mathbb{N}}$ with U_i open of M and V_i open of N, such that the U_i cover M and $F(U_i) \subseteq V_i$, so the V_i cover $F(M)$.
By virtue of Proposition 1.27, and by considering the local expression of F, $f := \phi_i \circ F \circ \psi_i^{-1}$, it sufficies to prove:

Theorem 1.30. Let U be an open in $\mathbb{R}^m$, $f : U \to \mathbb{R}^n$ be a smooth function, and let C be the set of critical points of f in U. Then $f(C)$ has measure zero in $\mathbb{R}^n$.

Note that the theorem is trivially true for $m = 0$, i.e, for the unique non-empty set $U = \{p\} = \mathbb{R}^0$, since $n \geq 1$ because we have defined the measure zero condition for this range of values and a point in $\mathbb{R}^n$ has measure zero. We will assume that the condition holds for $m - 1$ and prove it for m.
Let us define the sequence of nested subsets $C \supseteq C_1 \supseteq C_2 \supseteq \dots$ where C_i for $i \geq 1$ is defined as the set of all $x \in U$ such that the partial derivatives of f of order $\leq i$ are zero at x. The sets C_i are closed, since they are defined as the preimages of zero under the derivatives of f.
Note that:

$$C = (C - C_1) \cup (C_1 - C_2) \cup \dots \cup (C_{k-1} - C_k) \cup C_k$$

for any $k \in \mathbb{N}$, where the symbol $-$ denotes the set difference. In the following lemmas, we will prove that the images $f(C - C_1)$ and $f(C_k - C_{k+1})$ have measure zero for all $k \in \mathbb{N}$, and that there exists a sufficiently large k such that $f(C_k)$ has measure zero. This will prove that $f(C)$ has measure zero.

Lemma 1.31. The image $f(C - C_1)$ has measure zero.

Proof. Around each $x \in C - C_1$, we will find an open set U_x such that $f(C \cap U_x)$ has measure zero in $\mathbb{R}^n$. Since $C - C_1$ is covered by a countable collection of these open sets (by the Second Axiom of Countability), by virtue of Proposition 1.19, this will prove that $f(C - C_1)$ has measure zero.
By definition of C_1, for $x \in C - C_1$ there exists some partial derivative of f which is not zero at x. Without loss of generality, let us assume that $\frac{\partial f_1}{\partial x_1} \neq 0$ at x. Consider the function $h : U \to \mathbb{R}^m$ defined by:

$$h(x_1, \dots, x_m) := (f_1(x_1, \dots, x_m), x_2, \dots, x_m).$$

Since $d_x h$ is non-singular, by the Inverse Function Theorem there exists an open neighbourhood U_x of x and an open neighbourhood U'_x of $h(x)$ such that h is a diffeomorphism between U_x and U'_x.
Consider the composition $g := f \circ h^{-1} : U'_x \subseteq \mathbb{R}^m \to \mathbb{R}^n$ which has the same critical values as the restriction of f to U_x (since h^{-1} has no critical points). By definition of h, the function g maps points of the form $(t, x_2, \dots, x_m)$ into $(t, y_2, \dots, y_n)$, that is, the first coordinate of the point remains the same. Then, g induces a map $g^t : (\{t\} \times \mathbb{R}^{m-1}) \cap U'_x \to \{t\} \times \mathbb{R}^{n-1}$ for each fixed t. Observe that the differential of g has the expression:

$$\left(\frac{\partial g_i}{\partial x_j}\right) = \begin{pmatrix} 1 & 0 \\ * & \left(\frac{\partial g_i^t}{\partial x_j}\right) \end{pmatrix}.$$

Then, a point of $(\{t\} \times \mathbb{R}^{m-1}) \cap U'_x$ is critical for g if and only if is critical for g^t, since the differential of g does not have maximal rank if and only if the differential of g^t does not have maximal rank.
Note that since g^t is a smooth function defined on an open subset of $\mathbb{R}^{m-1}$, by induction hypothesis, the set of critical values of g^t, which is the same as for g, has measure zero in $V_t = \{t\} \times \mathbb{R}^{n-1}$ for each t.
Since the set of critical points of g is a closed set, by σ-compactness it can be expressed as a countable union of compact sets. Then, the image of the critical points by g, that is, the set of the critical values of g, can be expressed as a countable union of compact sets, and therefore is a closed set.
Since the set of critical values of g is a closed set that has measure zero in V_t for each t, by Fubini's Theorem 1.23 the set of critical values of g has measure zero in $\mathbb{R}^n$. Therefore, since the critical values of g are the same as those of f restricted to U_x, $f(C \cap U_x)$ has measure zero in $\mathbb{R}^n$. □

Lemma 1.32. The image $f(C_k - C_{k+1})$ has measure zero for $k \geq 1$.

Proof. We will follow a similar argument to that of the previous lemma. Around each $x \in C_k - C_{k+1}$, we will find an open set U_x such that $f(C_k \cap U_x)$ has measure zero in $\mathbb{R}^n$.
Let ρ be the k-derivative of f that is zero at a point $x \in C_k - C_{k+1}$ but has a first derivative $\frac{\partial \rho}{\partial x_1} \neq 0$. Consider the function $h : U \to \mathbb{R}^m$ defined by

$$h(x_1, \ldots, x_m) := (\rho(x_1, \ldots, x_m), x_2, \ldots, x_m).$$

which, by the Inverse Function Theorem, maps a neighbourhood U_x of x diffeomorphically onto an open set U'_x.
Note that by definition, the image $h(C_k \cap U_x)$ is contained in $\{0\} \times \mathbb{R}^{m-1}$. Then, if we consider the composition $g := f \circ h^{-1} : U'_x \subseteq \mathbb{R}^m \to \mathbb{R}^n$, which has the same critical values as the restriction of f to U_x, all the critical points of type C_k of g are contained in $\{0\} \times \mathbb{R}^{m-1}$.
Let us now consider the restriction of g to $\{0\} \times \mathbb{R}^{m-1}$, $g' : (\{0\} \times \mathbb{R}^{m-1}) \cap U'_x \to \mathbb{R}^n$. Since g' is a smooth function defined on an open subset of $\mathbb{R}^{m-1}$, by induction hypothesis, the set of critical values of g' has measure zero in $\mathbb{R}^n$.
In particular, the set of critical points of type C_k of g are the same as those of g', and therefore their image by g has measure zero in $\mathbb{R}^n$. Hence, since the critical

values of g are the same as those of f restricted to U_x, $f(C_k \cap U_x)$ has measure zero in $\mathbb{R}^n$. □

Lemma 1.33. The image $f(C_k)$ has measure zero for $k > \frac{m}{n} - 1$.

Proof. Let $S \subseteq \mathbb{R}^m$ be a cube with sides of length δ such that $\overline{S} \subseteq U$. We will prove that for a k sufficiently large, precisely $k > \frac{m}{n} - 1$, $f(C_k \cap S)$ has measure zero. Since C_k can be covered by a countable collection of cubes (by the Second Axiom of Countability), by virtue of the Proposition 1.19, this will prove that $f(C_k)$ has measure zero.
Let $x \in C_k \cap S$, $h \in \mathbb{R}^m$ such that $x + h \in S$. From Taylor's Theorem and the definition of C_k, we have:

$$f(x+h) = f(x) + R(x,h), \tag{1.3}$$

where R is a smooth function, well defined for the described x and h, with image in $\mathbb{R}^n$. Since $\overline{S}$ is compact, from Weierstrass' Theorem, the function R satisfies:

$$\|R(x,h)\| < a\|h\|^{k+1} \tag{1.4}$$

for $a \in \mathbb{R}$ a constant which depends only on f and S.
Let us subdivide S into r^m cubes of side δ/r. Let S_1 be the cube of the subdivision that contains x. Any point of S_1 can be described as $x + h$ with:

$$\|h\| < \frac{\delta}{r}\sqrt{m} \tag{1.5}$$

(as we did while proving Proposition 1.26). From (1.3), (1.4) and (1.5) we deduce that $f(S_1) \subseteq B(f(x), a(\delta\sqrt{m})^{k+1}\frac{1}{r^{k+1}})$. Hence, if $b = 2a(\delta\sqrt{m})^{k+1}$ we have that $f(S_1)$ is contained in a cube with side b/r^{k+1}.
Therefore, $f(C_k \cap S)$ is covered by, at most, r^m cubes, each of volume $(b/r^{k+1})^n$. This covering has total volume less than $r^m(\frac{b}{r^{k+1}})^n = b^n\frac{1}{r^{n(k+1)-m}}$. If $k > \frac{m}{n} - 1$, then for a sufficiently large r, the volume of the covering is smaller than an arbitrarily small $\varepsilon > 0$. Hence, $f(C_k \cap S)$ has measure zero. □

This completes the proof of Sard's Theorem. □

1.3 Abundance and Density of Morse Functions

In this section, we apply Whitney's Theorem and Sard's Theorem to demonstrate the abundance and density properties of Morse functions on a compact manifold. Specifically, we show that Morse functions can be defined with respect

to almost every point of the manifold, and that they form a dense subset of the space of smooth functions endowed with the C^∞ topology.

Throughout this unit (and the following ones as well), by virtue of Whitney's Theorem 1.12, we assume that any compact manifold M is embedded in the Euclidean space $\mathbb{R}^n$, so that points $p \in M$ are regarded as n-tuples in $\mathbb{R}^n$. We start by showing the abundance property of Morse functions on compact manifolds.

Proposition 1.34. (Existence and Abundance of Morse functions) Let M be a compact smooth manifold embedded in the Euclidean space $(\mathbb{R}^n, \|\cdot\|)$.
For almost every point $p \in \mathbb{R}^n$, the function $f_p \in C^\infty(M)$ defined below is a Morse function:

$$f_p(q) = \|q - p\|^2.$$

Proof. Since a Morse function is defined by the absence of degenerate critical points, let us first find the critical points of f_p for an arbitrary $p \in \mathbb{R}^n$.
Let $f_p(q) = (q_1 - p_1)^2 + \cdots + (q_n - p_n)^2$, where q is here its image by the embedding defined in M. Then, for every $q \in M$, we have the differential $d_q f_p : T_q M \to T_{f_p(q)}\mathbb{R} \cong \mathbb{R}$ defined as $d_q f_p(v) = 2(q - p) \cdot v$. Then, $q \in M$ is a critical point of f_p if and only if $(q - p) \cdot v = 0$ for every $v \in T_q M$, that is,

$$q - p \in (T_q M)^\perp. \tag{1.6}$$

Let $\psi : V \subseteq \mathbb{R}^m \to M$ be a local parametrization of M with coordinates $x = (x_1, \ldots, x_m)$. Let $\{\frac{\partial}{\partial x_i}\}_{i=1}^m$ be the coordinate basis of $T_x V$ and $\{\frac{\partial \psi}{\partial x_i}\}_{i=1}^m = \{d\psi(\frac{\partial}{\partial x_i})\}_{i=1}^m$ the coordinate basis of $T_{\psi(x)}M$. Let us use the abuse of notation $\psi(x_1, \ldots, x_m) = q(x_1, \ldots, x_m)$ for $q = \psi(x_1, \ldots, x_m)$.
In the local expression of f_p in these coordinates, we have $\frac{\partial f_p}{\partial x_i}(q) = d_q f_p(\frac{\partial q}{\partial x_i}) = 2(q - p) \cdot \frac{\partial q}{\partial x_i}$, and

$$\frac{\partial^2 f_p}{\partial x_j \partial x_i}(q) = 2\left(\frac{\partial q}{\partial x_j} \cdot \frac{\partial q}{\partial x_i} + (q - p) \cdot \frac{\partial^2 q}{\partial x_j \partial x_i}\right) \tag{1.7}$$

for $1 \leq i, j \leq m$. Therefore, $q \in M$ is a non-degenerate critical point of f_p if and only if (1.6) is satisfied and the Hessian from (1.7) is invertible.

Now, we want to prove that the set

$$P := \{p \in \mathbb{R}^n : f_p \text{ is not a Morse function}\}$$

has measure zero in $\mathbb{R}^n$. Note that this set P is the same as the set of points $p \in \mathbb{R}^n$ such that exists a $q \in M$ satisfying (1.6) and with a Hessian from (1.7)

non-invertible. To show this, we will prove that the points in P are critical values of a certain smooth function, which is a sufficient condition to have measure zero by the Sard's Theorem 1.16.
We will define such a function in the space

$$N := \{(q, v) \in M \times \mathbb{R}^n : v \perp T_q M\},$$

which is the orthogonal normal fiber bundle of M in $\mathbb{R}^n$. Note that N is contained in the product smooth manifold $M \times \mathbb{R}^n$ of dimension $m + n$. Also, it is a submanifold of this product manifold, as we prove next.

Lemma 1.35. The orthogonal normal fiber bundle N is a submanifold of $M \times \mathbb{R}^n$ with dimension n.

Proof. Let $d\varphi : \mathbb{R}^n \to \mathbb{R}^n$ be a parametrization of $\mathbb{R}^n$ (where we have identified the tangent space $T\mathbb{R}^n$ with $\mathbb{R}^n$), with coordinates $(a_1, \ldots, a_n)$. We choose φ such that the basis $\{\frac{\partial \varphi}{\partial a_i}\}_{i=1}^n = \{d\varphi(\frac{\partial}{\partial a_i})\}_{i=1}^n$ is orthonormal, with first m vectors $w_1, \ldots, w_m \in T_q M$ and the remaining $n - m$ vectors $v_1, \ldots, v_{n-m} \in (T_q M)^\perp$, which form an orthonormal basis of $(T_q M)^\perp$. Note that $\{\frac{\partial q}{\partial x_1}, \ldots, \frac{\partial q}{\partial x_m}, v_1, \ldots, v_{n-m}\} \subseteq T_q M \cup (T_q M)^\perp$ is a basis of $T\mathbb{R}^n \cong \mathbb{R}^n$ (since vectors from basis of $T_q M$ and $(T_q M)^\perp$ are orthogonal and therefore linearly independent).
Let us define the map $\phi : V \times \mathbb{R}^{n-m} \subseteq \mathbb{R}^n \to N$ as:

$$\phi(x, t) = \phi(x_1, \ldots, x_m, t_1, \ldots, t_{n-m}) = \left(q(x_1, \ldots, x_m), \sum_{i=1}^{n-m} t_i v_i \right).$$

which is a parametrization of N (since it is exhaustive, injective, continuous, and has a continuous inverse).
In order to prove that N is a submanifold of $M \times \mathbb{R}$, we have to see that the inclusion $i : N \to M \times \mathbb{R}^n$ is an embedding. Since in N we have the induced topology of $M \times \mathbb{R}^n$, it is sufficient to see that $d_{(q,v)} i$ is injective for an arbitrary $(q, v) \in N$. Note that, locally, i acts as:

$$(x, t) \in V \times \mathbb{R}^{n-m} \xrightarrow{\phi} (q(x), v(t)) \in N \xrightarrow{i} (q(x), v(t)) \in M \times \mathbb{R}^n \xrightarrow{\psi^{-1} \times Id} (x, v(t)) \in \mathbb{R}^m \times \mathbb{R}^n$$

with an injective differential:

$$\begin{pmatrix} Id_{m \times m} & 0 & \ldots & 0 \\ 0 & v_1 & \ldots & v_{n-m} \end{pmatrix}.$$

□

The following lemma completes the proof of the proposition.

Lemma 1.36. The set of critical values of the function $E : N \to \mathbb{R}^n$ defined below is P:

$$E(q, v) := q + v.$$

Proof. Let $p = E(q, v) = q + v$. In the local parametrization ϕ of N defined in Lemma 1.35, we can locally express E as $E(x, t) = q(x) + \sum_{i=1}^{n-m} t_i v_i$, which has a differential:

$$d_{(x,t)}E = \begin{pmatrix} \frac{\partial E}{\partial x_1} & \dots & \frac{\partial E}{\partial x_m} & \frac{\partial E}{\partial t_1} & \dots & \frac{\partial E}{\partial t_{n-m}} \end{pmatrix} \in M_n(\mathbb{R})$$

where $\frac{\partial E}{\partial x_i} = \frac{\partial q}{\partial x_i} + \sum_{k=1}^{n-m} t_k \frac{\partial v_k}{\partial x_i}$ and $\frac{\partial E}{\partial t_i} = v_i$.

Let us consider the matrix:

$$\begin{pmatrix} \frac{\partial q}{\partial x_1} & \dots & \frac{\partial q}{\partial x_m} & v_1 & \dots & v_{n-m} \end{pmatrix} \in M_n(\mathbb{R})$$

which is an invertible matrix since its columns are linearly independent. The non-invertibility of $d_{(x,t)}E$ is equivalent to the non-invertibility of

$$\begin{pmatrix} \frac{\partial q}{\partial x_1} & \dots & \frac{\partial q}{\partial x_m} & v_1 & \dots & v_{n-m} \end{pmatrix}^t \circ d_{(x,t)}E = \begin{pmatrix} \left(\frac{\partial q}{\partial x_i} \cdot \frac{\partial E}{\partial x_j}\right)_{\substack{1 \le i \le m \\ 1 \le j \le m}} & \left(\frac{\partial q}{\partial x_i} \cdot \frac{\partial E}{\partial t_j}\right)_{\substack{1 \le i \le m \\ 1 \le j \le n-m}} \\ \left(v_i \cdot \frac{\partial E}{\partial x_j}\right)_{\substack{1 \le i \le n-m \\ 1 \le j \le m}} & \left(v_i \cdot \frac{\partial E}{\partial t_j}\right)_{\substack{1 \le i \le n-m \\ 1 \le j \le n-m}} \end{pmatrix}$$

which is equal to:

$$\begin{pmatrix} \left(\frac{\partial q}{\partial x_i} \cdot \frac{\partial q}{\partial x_j} + \sum_{k=1}^{n-m} t_k \frac{\partial q}{\partial x_i} \cdot \frac{\partial v_k}{\partial x_j}\right) & 0 \\ \left(\sum_{k=1}^{n-m} t_k v_i \cdot \frac{\partial v_k}{\partial x_j}\right) & Id \end{pmatrix}.$$

Then, the non-invertibility of $d_{(x,t)}E$ is equivalent to the non-invertibility of

$$\left(\frac{\partial q}{\partial x_i} \cdot \frac{\partial q}{\partial x_j} + \sum_{k=1}^{n-m} t_k \frac{\partial q}{\partial x_i} \cdot \frac{\partial v_k}{\partial x_j}\right) \in M_m(\mathbb{R}).$$

Since $\frac{\partial q}{\partial x_i} \cdot v_k = 0$, we have $0 = \frac{\partial}{\partial x_j}(\frac{\partial q}{\partial x_i} \cdot v_k) = \frac{\partial^2 q}{\partial x_j \partial x_i} \cdot v_k + \frac{\partial q}{\partial x_i} \cdot \frac{\partial v_k}{\partial x_j}$, so we can rewrite the last matrix as:

$$\left(\frac{\partial q}{\partial x_i} \cdot \frac{\partial q}{\partial x_j} - v \cdot \frac{\partial^2 q}{\partial x_j \partial x_i}\right). \tag{1.8}$$

Let us suppose that p is a critical value of E, that is, $d_{(x,t)}E$ is non-invertible . Note that $p - q = v \in (T_qM)^{\perp}$ (condition (1.6)) is satisfied by construction of N. Also, the matrix (1.8) can be expressed as:

$$\left(\frac{\partial q}{\partial x_i} \cdot \frac{\partial q}{\partial x_j} + (q-p) \cdot \frac{\partial^2 q}{\partial x_j \partial x_i}\right).$$

Since this matrix is non-invertible, the Hessian from (1.7) is non-invertible, which implies $p \in P$. □

This completes the proof of the proposition. □

Now we prove the mentioned density property of Morse functions.

Proposition 1.37. (Density of Morse functions) Let M be a compact smooth manifold. Then, Morse functions are dense in $C^\infty(M)$ with respect to the C^∞ topology. That is, any smooth function in $C^\infty(M)$ can be approximated arbitrarily closely in the C^∞ topology by Morse functions.

Proof. Let $\mathcal{M}(M) \subseteq C^\infty(M)$ be the set of Morse functions in M. We have to prove that for every $f \in C^\infty(M)$, there exists a $g \in \mathcal{M}(M)$ such that $\sup_{q\in M} \|f^{(k)}(q) - g^{(k)}(q)\| < \varepsilon$ for every $\varepsilon > 0$, where $^{(k)}$ denotes the k-derivative of the function.
Let us fix a real number $c \in \mathbb{R} - \{0\}$. Let n be sufficiently large such that $h : M \to \mathbb{R}^n$ defined as

$$h(q) := (f(q), h_2(q), \dots h_n(q)),$$

where $h_i \in C^\infty(M)$, is an embedding of M in $\mathbb{R}^n$. This is possible since if $h' : M \to \mathbb{R}^{n-1}$ is an embedding defined as $h'(q) = (h_2(q), \dots h_n(q))$, then h is also an embedding (it is an immersion because the rank of the differential remains the maximum possible rank $m \leq n-1$, and it is homeomorphic to its image).
Let $p = (-c + \varepsilon_1, \varepsilon_2, \dots, \varepsilon_n) \in \mathbb{R}^n$ be a perturbation of $(-c, 0, \dots, 0)$ such that the function f_p, defined as

$$f_p(q) = \|h(q) - (-c + \varepsilon_1, \varepsilon_2, \dots, \varepsilon_n)\|^2,$$

is a Morse function (according to Proposition 1.34). Then,

$$g(q) := \frac{f_p(q) - c^2}{2c}$$

is also a Morse function. Note that for every $q \in M$, g satisfies:

$$g(q) = \frac{1}{2c}\left((f(q)+c-\varepsilon_1)^2 + (h_2(q)-\varepsilon_2)^2 + \cdots + (h_n(q)-\varepsilon_n)^2 - c^2\right) =$$

$$= f(q) + \frac{1}{c}[\frac{1}{2}(f(q)^2 + \sum_i h_i(q)^2) - \varepsilon_1 f(q) - \sum_i \varepsilon_i h_i(q)] + \sum \varepsilon_i^2 - \varepsilon_1 =$$

$$= f(q) + \frac{1}{c} l(q, \varepsilon_i) + \sum \varepsilon_i^2 - \varepsilon_1,$$

where $l \in C^\infty(M)$.

Therefore, for every $\varepsilon > 0$, there exists $c > 0$ large enough and $\varepsilon_1, \ldots, \varepsilon_n$ small enough such that $g \in \mathcal{M}(M)$ and the following inequalities hold for every $q \in M$:

$$\|g(q) - f(q)\| \leq \frac{1}{c} C^0(\varepsilon_i) + \sum_i \varepsilon_i^2 + \varepsilon_1 < \varepsilon,$$

$$\|g^{(k)}(q) - f^{(k)}(q)\| \leq \frac{1}{c} C^k(\varepsilon_i) < \varepsilon,$$

where we have used Weierstrass' Theorem to bound $l^{(k)} \in C^\infty(M)$ by $C^k(\varepsilon_i) \in \mathbb{R}^+$ in the compact M.

□

Chapter 2

Morse Theory

In this chapter, we present and demonstrate the main results of Morse theory. We start by introducing Morse's Lemma, which describes the local expression of a smooth function in a neighbourhood of a non-degenerate critical point. Next, we describe the topological structure of manifolds in terms of the subsets M^a for an arbitrary Morse function, studying the key role critical points play in describing the homotopy type of these subsets. Furthermore, we also study the topological structure of compact manifolds, analysing how the critical points of a Morse function describe a cell complex structure that is homotopy equivalent to the underlying compact manifold. We conclude the chapter by illustrating the results and discussing their applications.

2.1 Morse's Lemma

In this section, we introduce and prove Morse's Lemma, which states that the behaviour of a Morse function at a non-degenerate critical point can be completely characterized by the index of the function at that point. More precisely:

Lemma 2.1. (Morse's Lemma) Let M be a manifold, $f \in C^\infty(M)$ and $p \in M$ be a non-degenerate critical point of f.
There exists a local coordinate system $(y_1, \ldots, y_m)$ centred at p such that the local expression of f in that system is

$$f(y_1, \ldots, y_m) = f(0) - y_1^2 - \cdots - y_\lambda^2 + y_{\lambda+1}^2 + \cdots + y_m^2$$

where λ is the index of f at p.

Proof. First, let us prove that if such an expression for f exists, then λ is the index of f at p.

Let $(z_1, \ldots, z_m)$ be an arbitrary coordinate system that satisfies the conditions of the lemma. Then, it has a Hessian with an expression:

$$\frac{\partial^2 f}{\partial z_i \partial z_j}(0) = \begin{cases} -2, & \text{if } 1 \leq i = j \leq \lambda, \\ +2, & \text{if } \lambda + 1 \leq i = j \leq m, \\ 0, & \text{otherwise.} \end{cases}$$

Note how there is a subspace of T_pM of dimension λ where the Hessian is negative definite, and a subspace V of dimension $m - \lambda$ where it is positive definite. If there were a subspace of T_pM of dimension greater than λ on which the Hessian were negative definite, then it would intersect V, which is impossible. Therefore λ is the index of f at p.

Now we will see that the local coordinate system $(y_1, \ldots, y_m)$, which satisfies the conditions of the lemma exists. First, let us consider the following lemma:

Lemma 2.2. Let $f : V \subseteq \mathbb{R}^n \to \mathbb{R}$ be a smooth function defined in a convex neighbourhood V of 0 in $\mathbb{R}^n$, with $f(0) = 0$. Then,

$$f(x_1, \ldots, x_n) = \sum_{i=1}^{n} x_i g_i(x_1, \ldots, x_n)$$

for some smooth functions $g_i : V \to \mathbb{R}$ with $g_i(0) = \frac{\partial f}{\partial x_i}(0)$.

Proof. We can express f as:

$$f(x_1, \ldots, x_n) = \int_0^1 \frac{d}{dt} f(tx_1, \ldots, tx_n)\, dt = \int_0^1 \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(tx_1, \ldots, tx_n) \cdot x_i\, dt.$$

Since f is smooth, the sum switches with the integral, and we let

$$g_i(x_1, \ldots, x_n) := \int_0^1 \frac{\partial f}{\partial x_i}(tx_1, \ldots, tx_n)\, dt.$$

□

Let us assume that p has the zero vector $0 \in \mathbb{R}^m$ as coordinates and that $f(p) = f(0) = 0$ (equivalently, the following argument can be applied to $\tilde{f} := f - f(p)$). By applying Lemma 2.2 to the local expression of f, we can write $f(x_1, \ldots, x_m) = \sum_{j=1}^{m} x_j g_j(x_1, \ldots, x_m)$ for $(x_1, \ldots, x_m)$ a coordinate system defined in some convex neighbourhood of 0.
Note that since 0 is a critical point, we have $g_j(0) = \frac{\partial f}{\partial x_j}(0) = 0$. We can therefore

apply again Lemma 2.2, this time to the g_j, to obtain $g_j(x_1, \ldots, x_m) = \sum_{i=1}^m x_i h_{ij}(x_1, \ldots, x_m)$ for certain smooth functions h_{ij}. It follows that

$$f(x_1, \ldots, x_m) = \sum_{i,j=1}^m x_i x_j h_{ij}(x_1, \ldots, x_m).$$

Note that we can assume that $h_{ij} = h_{ji}$. Indeed, if we define $\bar{h}_{ij} := \frac{1}{2}(h_{ij} + h_{ji})$ which satisfies $\bar{h}_{ij} = \bar{h}_{ji}$, then we can write $f = \sum_{i,j} x_i x_j \bar{h}_{ij}$. In addition, the matrix $(\bar{h}_{ij}(0))$ is equal to $\left(\frac{1}{2}\frac{\partial^2 f}{\partial x^i \partial x^j}(0)\right)$, and then it is non-singular since p is non-degenerate.

Suppose by induction that there exist coordinates $(u_1, \ldots, u_m)$ in a neighbourhood U_1 of 0 such that

$$f = \sum_{i \leq r-1} \pm (u_i)^2 + \sum_{i,j \geq r}^m u_i u_j H_{ij}$$

throughout U_1, where $H_{ij} = H_{ij}(u_1, \ldots, u_m)$ are smooth functions in U_1 such that $H_{ij} = H_{ji}$. Note that the case $r = 1$ is satisfied for the coordinates $(x_1, \ldots, x_m)$ and the functions h_{ij} (base case of induction).

After a linear change in the last $m - r + 1$ coordinates we may assume that $H_{rr}(0) \neq 0$. Now, let $g = g(u_1, \ldots, u_m)$ denote the square root of $|H_{rr}(u_1, \ldots, u_m)|$, which is a smooth non-zero function throughout some smaller neighbourhood $U_2 \subseteq U_1$ of 0. Note that for the case $r = 1$ this is true because $(h_{ij}(0))$ is non-singular since p is non-degenerate.

Let us introduce new coordinates $(v_1, \ldots, v_m)$ by $v_i := u_i$ for $i \neq r$, and $v_r := g\left[u_r + \frac{1}{H_{rr}} \sum_{i>r}^m u_i H_{ir}\right]$. Applying the Inverse Function Theorem, we have that $(v_1, \ldots, v_m)$ form a coordinate system within some sufficiently small neighbourhood U_3 of 0.

Now, since:

$$v_r^2 = |H_{rr}| \left[u_r^2 + \frac{2u_r}{H_{rr}} \sum_{i>r}^m u_i H_{ir} + \frac{1}{H_{rr}^2}\left(\sum_{i>r}^m u_i H_{ir}\right)^2\right] =$$

$$= \pm u_r^2 H_{rr} \pm \sum_{i>r}^m u_i u_r H_{ir} \pm \sum_{i>r}^m u_r u_i H_{ri} \pm \frac{1}{H_{rr}}\left(\sum_{i>r}^m u_i H_{ir}\right)^2$$

we can rewrite f as

$$f = \sum_{i \leq r} \pm (v_i)^2 + \sum_{i,j \geq r+1}^m v_i v_j H'_{ij}$$

throughout U_3, where $H'_{ij} = H'_{ij}(v_1, \ldots, v_m)$ are smooth functions in U_3 such that $H'_{ij} = H'_{ji}$. This completes the induction, which implies that exists a coordinate system $(y_1, \ldots, y_m)$ (for the step $'r = m'$) which satisfies the conditions of the lemma.

□

Observe that, according to the expression of f given in the lemma, the non-degenerate critical point p is the only critical point in the neighbourhood U. This immediately implies:

Corollary 2.3. Non-degenerate critical points of a smooth function are isolated.

Furthermore, we have the following result:

Corollary 2.4. If M is a compact manifold, a Morse function $f \in C^\infty(M)$ has finitely many critical points.

Proof. Since f is a Morse function, all critical points of f are non-degenerate and, by the previous corollary, are also isolated. Then, if C is the set of critical points of f, there exists a covering of C by open sets such that each open set contains one and only one critical point. Since C is closed for being the preimage of zero of the differential of f, and C is contained in the compact M, then C is compact and therefore there exists a finite subcover of C, which implies that C is finite. □

2.2 First Theorem of Morse

In this section, we introduce and prove what we refer to as the First Theorem of Morse. This result establishes diffeomorphisms between subsets M^a and M^b of M, provided that the region between them contains no critical points.

To prove the theorem, we first present the following result concerning vector fields.

Lemma 2.5. Let M be a manifold and let $X \in \mathcal{X}(M)$ be a vector field. If X vanishes outside of a compact subset K of M, then it generates a unique uniparametric group of diffeomorphisms of M.

Proof. Our aim is to find a unique uniparametric group of diffeomorphisms, that is, a smooth function $\phi : \mathbb{R} \times M \to M$ which satisfies:

(I) For each $t \in \mathbb{R}$, the map $\phi_t : M \to M$, where t is fixed in ϕ, is a diffeomorphism.

(II) For all $t, s \in \mathbb{R}$ we have $\phi_{t+s} = \phi_t \circ \phi_s$.

We know from the differential geometry course taught by Dr Ricardo García López, that for each $p \in M$ there exists a unique integral curve of X by p which is maximal. Let us denote this curve by $\phi_p : I_p \to M$, where $I_p \subseteq \mathbb{R}$ is the maximal interval of definition. The integral curve satisfies:

$$\begin{cases} \frac{d\phi_p(t)}{dt} = X_{\phi_p(t)} \\ \phi_p(0) = p \end{cases} \tag{2.1}$$

for each $t \in I_p$. The maximality and uniqueness means that if $\beta : J \to M$ is another integral curve of X by p, then the interval $J \subseteq I_p$ and $\beta = \phi_p|_J$.
We may denote $\phi_p(t) = \phi(t, p) = \phi_t(p)$ for each $p \in M$ and $t \in I_p$. The set $\Omega := \bigcup_{p \in M} I_p \times \{p\}$ is an open subset of $\mathbb{R} \times M$, and the function $\phi : \Omega \to M$ is smooth in both variables. This function is called the maximal flow of X.
Another result of the differential geometry course is that for each $p \in M$ and for a fixed $t \in I_p$ we have that $I_{\phi_p(t)} = I_p - t$ and $\phi_{s+t}(p) = (\phi_s \circ \phi_t)(p)$ for $s \in I_p - t$.
Then, we can take for each point of M a neighbourhood U and a number $\varepsilon > 0$ such that equation (2.1) has a unique (smooth) solution for $p \in U$ and $t \in (-\varepsilon, \varepsilon)$.
Since K is a compact subset of M, it can be covered by a finite number of such neighbourhoods , say $\{U_i\}_{i=1}^k$. Then, we can consider the value $\varepsilon_0 := \min_{1 \leq i \leq k} \varepsilon_i$ such that the integral curves $\{\phi_p\}_{p \in K}$ are all defined for $|t| < \varepsilon_0$.
Since $\phi_t(p) \equiv p$ are the solutions for $p \notin K$ (where $X = 0$), which are defined for all $t \in \mathbb{R}$, we have that $\{\phi_t(p)\}$ are integral curves of X that, for each $p \in M$, are defined in the interval $(-\varepsilon_0, \varepsilon_0)$. These functions are smooth in both variables and satisfy $\phi_{t+s} = \phi_t \circ \phi_s$ for $t, s, t+s \in (-\varepsilon_0, \varepsilon_0)$. Since for each $t \in (-\varepsilon_0, \varepsilon_0)$ we have $-t \in (-\varepsilon_0, \varepsilon_0)$ and $\phi_t \circ \phi_{-t} = \phi_0 = Id$, the functions are diffeomorphisms of M onto itself.

To define the maps ϕ_t for $|t| \geq \varepsilon_0$ we express t as a multiple of $\varepsilon_0/2$ plus a remainder r with $|r| < \varepsilon_0/2$. For $t = k(\varepsilon_o/2) + r$ with $k \geq 0$, we define:

$$\phi_t := \phi_{\varepsilon_0/2} \circ \overset{k)}{\cdots} \circ \phi_{\varepsilon_0/2} \circ \phi_r$$

and for $k < 0$ we do the same but replacing $\phi_{\varepsilon_0/2}$ by $\phi_{-\varepsilon_0/2}$ and we iterate it $-k$ times.
Since the ϕ_t for $|t| \geq \varepsilon_0$ are compositions of $\phi_{t'}$ with $|t'| < \varepsilon_0$, we finally have that the ϕ_t are, for all $t \in \mathbb{R}$, well defined, smooth, and satisfy $\phi_{t+s} = \phi_t \circ \phi_s$, so they are diffeomorphisms and form a uniparametric group of diffeomorphisms generated by X. The uniqueness follows from the uniqueness of the solutions stated above and the procedure that we followed. $\square$

We now introduce the main result of this section.

Theorem 2.6. (First Theorem of Morse) Let M be a manifold and let $f \in C^\infty(M)$. Let $a < b$ be real values, and suppose that $f^{-1}[a,b]$ is compact and has no critical points of f. Then, M^a is diffeomorphic to M^b. Furthermore, M^a is a deformation retract of M^b, so that the inclusion map $M^a \hookrightarrow M^b$ is a homotopy equivalence.

The idea of the proof is to endow M with a Riemannian metric and push M^b down to M^a along the orthogonal trajectories of the hypersurfaces $f = c$, where $c \in [a,b]$ are constants. At every point of these hypersurfaces, the orthogonal direction is given by the gradient of the function at that point, which we now define.

Definition 2.7. Let $(M, \langle \cdot, \cdot \rangle)$ be a Riemannian manifold, and let $f \in C^\infty(M)$. We define the **gradient** of f as the vector field $\operatorname{grad} f \in \mathcal{X}(M)$ characterized by the identity:

$$\langle X, \operatorname{grad} f \rangle = X(f)$$

for every $X \in \mathcal{X}(M)$. We say $X(f) \in C^\infty(M)$ is the directional derivative of f in the direction of X.

Proposition 2.8. For a local coordinate system $(x_1, \ldots, x_m)$ of M, the local expression of $\operatorname{grad} f$ has components

$$a^i = \sum_{j=1}^{m} g^{ij} \frac{\partial f}{\partial x_j}$$

for $1 \leq i \leq m$, where g^{ij} are the components of the inverse of the metric $g = \langle\ \rangle : \mathcal{X}(M) \times \mathcal{X}(M) \to C^\infty(M)$.

Proof. Let $\operatorname{grad} f = a^k \frac{\partial}{\partial x_k}$, where $a^k \in C^\infty(M)$, be the local expression of $\operatorname{grad} f$, where we use Einstein's notation. Let $X \in \mathcal{X}(M)$, with local expression $X = b^j \frac{\partial}{\partial x_j}$, where $b^j \in C^\infty(M)$. Then:

$$0 = \langle X, \operatorname{grad} f \rangle - X(f) = g(b^j \frac{\partial}{\partial x_j}, a^k \frac{\partial}{\partial x_k}) - b^j \frac{\partial f}{\partial x_j} = (g_{jk} a^k - \frac{\partial f}{\partial x_j}) b^j$$

which implies, for each j, that $g_{jk} a^k = \frac{\partial f}{\partial x_j}$. Multiplying by a row i of the inverse of g we get $g^{ij} g_{jk} a^k = \delta^i_k a^k = a^i = g^{ij} \frac{\partial f}{\partial x_j}$. □

Now, let us give a proof of the First Theorem of Morse.

Proof of Theorem 2.6. Let us choose a Riemannian metric on M, and let $\langle \cdot\,,\cdot \rangle$ denote the inner product determined by this metric.
Let V be an open of M which contains the compact $C := f^{-1}[a,b]$. Let $\gamma : M \to [0,1]$ be the plateau function associated to (V, C), which we presented in the differential geometry course taught by Dr. Ricardo García López. It is characterized by the properties $\gamma|_C = 1$ and $C \subseteq \operatorname{supp}\gamma \subseteq V$ where $\operatorname{supp}\gamma$, which is the closure in M of the set $\{p \in M : \gamma(p) \neq 0\}$, is compact.
Let us define the smooth function associated with the plateau function, $\rho \in C^\infty(M)$, as:

$$\rho := \begin{cases} \langle \operatorname{grad} f, \operatorname{grad} f \rangle^{-1} \cdot \gamma, & p \in V, \\ 0, & p \notin V. \end{cases}$$

Note that this function is well defined because $\operatorname{grad} f = 0$ only at the critical points of f, and C has no critical points of f by hypothesis (and we can always take a neighbourhood V without critical points).
Then, we can define a vector field $X \in \mathcal{X}(M)$ as:

$$X := \rho \operatorname{grad} f.$$

This vector field vanishes outside the compact neighbourhood $\operatorname{supp}\gamma$ of C, and therefore, by Lemma 2.5, generates a unique uniparametric group of diffeomorphisms $\varphi : \mathbb{R} \times M \to M$.
For $p \in M$, let us consider the smooth function $t \mapsto f(\varphi_p(t))$ and also the differential field $\frac{d\varphi(t)}{dt} : p \mapsto \frac{d\varphi_p(t)}{dt} = d\varphi_p(t)(\frac{d}{dt})$. By expression (2.1) we have:

$$X(f)(\varphi_p(t)) = \frac{d\varphi(t)}{dt}(f)(p) = d\varphi(t)(\frac{d}{dt})(f)(p) = \frac{d}{dt}(f \circ \varphi(t))(p) = \frac{d}{dt}(f(\varphi_p(t))).$$

Now, if $\varphi_p(t)$ lies in the set $C = f^{-1}[a,b]$, we have $X_{\varphi_p(t)} = \frac{\operatorname{grad}_{\varphi_p(t)} f}{\langle \operatorname{grad}_{\varphi_p(t)} f, \operatorname{grad}_{\varphi_p(t)} f \rangle}$ and applying Definition 2.7 we obtain:

$$\frac{d}{dt}(f(\varphi_p(t)) = X(f)(\varphi_p(t)) = \langle X, \operatorname{grad} f \rangle(\varphi_p(t)) = 1,$$

which implies

$$f(\varphi_t(p)) = t + f(p). \tag{2.2}$$

Now, consider the diffeomorphism $\varphi_{b-a} : M \to M$. Since for $\varphi_{b-a}(p) \in f^{-1}[a,b]$ the identity (2.2) holds, for $p \in M^a$ we have $f(\varphi_{b-a}(p)) \leq b - a + a = b$. This implies that φ_{b-a} carries M^a diffeomorphically onto M^b.

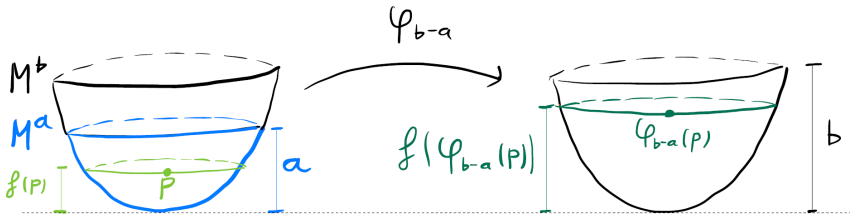

Figure 2.1: Illustration of the diffeomorphism φ_{b-a} acting on a point $p \in M^a$ and mapping it to M^b, for a height function f on a paraboloid. Observe that $f(\varphi_{b-a}(p)) = b - a + f(p)$.

Now, let us define a (smooth) uniparametric family of maps $r_t : M^b \to M^b$ by

$$r_t(p) = \begin{cases} p & \text{if } f(p) \leq a, \\ \varphi_{t(a-f(p))}(p) & \text{if } a \leq f(p) \leq b. \end{cases}$$

Note that $r_0 = id_{M^b}$, and r_1 defines a deformation retraction of M^b onto M^a.
Indeed, first note that $r_1 : M^b \to M^a$ since $r_1|_{M^a} = id_{M^a}$ by definition of r_1, and for $p \in f^{-1}[a,b]$ we have (applying expression (2.2)) that $f(r_1(p)) = f(\varphi_{a-f(p)}(p)) = a$ which implies $r_1(f^{-1}[a,b]) = f^{-1}(a) \subseteq M^a$.
Since for $i : M^a \hookrightarrow M^b$ we have $r_1 \circ i = r_1|_{M^a} = id_{M^a}$ and $i \circ r_1 = r_1 \sim r_0 = id_{M^b}$, where $\sim$ denotes the homotopy of functions given by the homotopy r_t, this proves that M^a is a deformation retract of M^b, with i as a homotopy equivalence. $\square$

2.3 Second Theorem of Morse

In this section, we introduce and demonstrate what we refer to as the Second Theorem of Morse. This fundamental result describes the structure of the sets M^a in the presence of non-degenerate critical points. Specifically, together with the First Theorem of Morse, it establishes that if the difference between sets M^a and M^b, with $a < b$, contains critical points, all of them non-degenerate, then M^b has the homotopy type of M^a with one cell attached for each critical point in the difference, where the dimension of each cell equals the index of the corresponding critical point.

First, we introduce some basic definitions.

Definition 2.9. For some $k \in \mathbb{N}$, we define the **k-cell** e^k as the closed k-dimensional disk in the Euclidean k-space:

$$e^k := D^k = \{x \in \mathbb{R}^k : \|x\| \leq 1\}.$$

The **boundary** of the k**-cell** is the $(k-1)$-sphere:

$$\dot{e}^k := S^{k-1} = \{x \in \mathbb{R}^k : \|x\| = 1\}.$$

We refer to spaces that are homeomorphic to e^k and $\dot{e}^k$ also as cells and boundaries of cells, respectively. By convention, e^0 is a point and $\dot{e}^0$ is the empty set.

Definition 2.10. Let X be a topological space, $k \in \mathbb{N}$. Let $g : \dot{e}^k \to X$ be a continuous map. Then, the space obtained by **attaching a** k**-cell** to X via g is defined as:

$$X \cup_g e^k := (X \sqcup e^k)/\sim_g$$

which is obtained by first taking the topological sum (i.e disjoint union) of X and e^k, and then identifying by the equivalence relation $\sim_g$ each $u \in \dot{e}^k$ with its image $g(u) \in X$.

These constructions allow us to state the Second Theorem of Morse.

Theorem 2.11. (Second Theorem of Morse) Let M be a manifold, $f \in C^\infty(M)$ and let $p \in M$ be a non-degenerate critical point of f with index λ. Let $f(p) = c$ and suppose there exists a $\varepsilon > 0$ such that $f^{-1}[c-\varepsilon, c+\varepsilon]$ is compact and contains no critical points of f other than p. Then, for all sufficiently small ε, the set $M^{c+\varepsilon}$ has the homotopy type of $M^{c-\varepsilon}$ with a λ-cell attached.

Let us first give an idea of the proof. In the demonstration, we will introduce a smooth function $F \in C^\infty(M)$ which coincides with f except in a small neighbourhood of p, where $F < f$. Thus, the region $F^{-1}(-\infty, c-\varepsilon]$ will consist of $M^{c-\varepsilon}$ together with a region H which contains p:

$$F^{-1}(-\infty, c-\varepsilon] = M^{c-\varepsilon} \cup H.$$

We will choose a suitable λ-cell, $e^\lambda \subseteq H$, such that $p \in e^\lambda$ and e^λ is a deformation retract of H. We will divide the proof into two parts. In the first part, we will show that $F^{-1}(-\infty, c+\varepsilon] = M^{c+\varepsilon}$ and we will apply the First Theorem of Morse to the region $F^{-1}[c-\varepsilon, c+\varepsilon]$ to prove that $M^{c-\varepsilon} \cup H$ is a deformation retract of $F^{-1}(-\infty, c+\varepsilon]$. In the second part, we will show that $M^{c-\varepsilon} \cup e^\lambda$ is a deformation retract of $M^{c-\varepsilon} \cup H$, which will complete the proof.

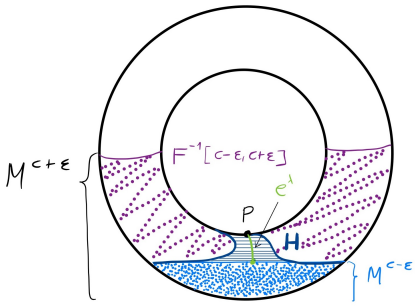

Figure 2.2: Schematic illustration of the described situation for the height function in a torus. The horizontal lines in H (dark blue region) represent the directions of the homotopy with the λ-cell e^λ (green curve). The region $M^{c-\varepsilon}$ (light blue shaded region), together with H, is a deformation retract of $M^{c+\varepsilon}$ (which is $M^{c-\varepsilon} \cup H$ together with the dotted purple region $F^{-1}[c-\varepsilon, c+\varepsilon]$).

Proof of Theorem 2.11. Let us choose a coordinate system $(x_1, \dots, x_m)$ in a neighbourhood U of p so that the identity $f = c - x_1^2 - \cdots - x_\lambda^2 + x_{\lambda+1}^2 + \cdots + x_m^2$ holds in U (according to Morse's Lemma 2.1). Thus, the critical point p will have coordinates $x_1(p) = \cdots = x_m(p) = 0$, and it is the only critical point in U, as follows from the local expression of f.
Let us define two functions $\xi, \eta : U \longrightarrow [0, \infty)$ with local expressions $\xi := x_1^2 + \cdots + x_\lambda^2$, and $\eta := x_{\lambda+1}^2 + \cdots + x_m^2$. Then, we can express $f = c - \xi + \eta$.

Let $\varepsilon > 0$ be sufficiently small so that the region $f^{-1}[c-\varepsilon, c+\varepsilon]$ is compact and contains no critical points of f other than p, and the image of U under the chart

$$(x_1, \dots, x_m) : U \longrightarrow V \subseteq \mathbb{R}^m$$

contains the disk

$$D =: \{(x_1, \dots, x_m) : \sum_{i=1}^m x_i^2 \leq 2\varepsilon\} = \{\xi + \eta \leq 2\varepsilon\}.$$

Let us consider the cell

$$e^\lambda := \{(x_1, \dots, x_m) : x_1^2 + \cdots + x_\lambda^2 \leq \varepsilon,\ x_{\lambda+1} = \cdots = x_m = 0\} = \{\xi \leq \varepsilon,\ \eta = 0\}.$$

Note the identities in U:

$$M^{c-\varepsilon} = \{\xi \geq \eta + \varepsilon\}, \quad M^{c+\varepsilon} = \{\eta \leq \xi + \varepsilon\}.$$

Also note that $e^\lambda \cap M^{c-\varepsilon}$ is precisely the boundary $\dot{e}^\lambda = \{\xi = \varepsilon,\ \eta = 0\}$, so that e^λ is attached to $M^{c-\varepsilon}$ as required.

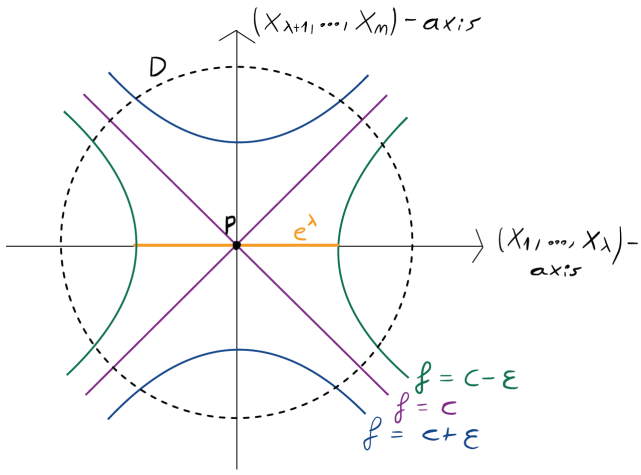

Figure 2.3: Schematic illustration of the described sets of U with respect to the planes $x_{\lambda+1} = \cdots = x^m = 0$ (horizontal direction) and $x_1 = \cdots = x_\lambda = 0$ (vertical direction). The dashed circle represents the boundary of D. The orange horizontal line represents the cell e^λ and the purple lines represent $f^{-1}(c)$. The green and blue hyperbolas represent the hypersurfaces $f^{-1}(c-\varepsilon)$ and $f^{-1}(c+\varepsilon)$, respectively. The region between the two blue hyperbolas represents the set $M^{c+\varepsilon}$, the region between the two green hyperbolas represents the complement of $M^{c-\varepsilon}$, and the region enclosed by all four hyperbolas corresponds to $f^{-1}[c-\varepsilon, c+\varepsilon]$.

Now, we will construct a new smooth function $F : M \longrightarrow \mathbb{R}$ as follows. Let $\mu : \mathbb{R} \longrightarrow \mathbb{R}$ be a C^∞ function satisfying the conditions:

$$\begin{aligned} \mu(0) &> \varepsilon, \\ \mu(r) &= 0, \quad \text{for } r \geq 2\varepsilon, \\ -1 < \mu'(r) &\leq 0, \quad \text{for all } r, \end{aligned}$$

where $\mu'(r) = \frac{d\mu}{dr}$.
Let F coincide with f outside the coordinate neighbourhood U, and let

$$F := f - \mu\left(\xi + 2\eta\right)$$

within this coordinate neighbourhood. The function F is a well defined smooth function throughout M. Indeed, let

$$E =: \{(x_1, \dots, x_m) : \sum_{i=1}^{\lambda} x_i^2 + 2 \sum_{i=\lambda+1}^{m} x_i^2 \leq 2\varepsilon\} = \{\xi + 2\eta \leq 2\varepsilon\}.$$

Since $\xi + \eta \leq \xi + 2\eta$, we have $E \subseteq D \subseteq V$. Thus, F is well defined in U. The function μ takes its maximum value at p, thus F takes its minumum value at p, where $F(p) < c - \varepsilon$. The function μ decreases (more precisely, it is non-increasing)

smoothly as we move across the ellipsoids contained in E, from the point p to the boundary of E. Consequently, the function F increases smoothly in these paths, until μ becomes zero outside the interior of E, and therefore F becomes f.

Let us structure by five steps the proof of the theorem. Steps three and five together prove the result.

Step 1: The region $F^{-1}(-\infty, c+\varepsilon]$ coincides with the region $M^{c+\varepsilon}$.

Indeed, since $F \leq f$ in M, trivially we have $M^{c+\varepsilon} \subseteq F^{-1}(-\infty, c+\varepsilon]$. On the other hand, if $F \leq c+\varepsilon$, outside E we have $f = F \leq c+\varepsilon$ and in E we have

$$f = c - \xi + \eta \leq c + \frac{1}{2}\xi + \eta \leq c + \varepsilon,$$

which proves $F^{-1}(-\infty, c+\varepsilon] \subseteq M^{c+\varepsilon}$.

Step 2: The critical points of F are the same as those of f.

Indeed, since $F = f$ outside U, and in U the only critical point of f is p, we have to see that the only critical point of F in U is p.
Note that in U, we have

$$\frac{\partial F}{\partial \xi} = -1 - \mu'(\xi + 2\eta) < 0, \quad \frac{\partial F}{\partial \eta} = 1 - 2\mu'(\xi + 2\eta) \geq 1.$$

Since $dF = \frac{\partial F}{\partial \xi}\, d\xi + \frac{\partial F}{\partial \eta}\, d\eta$, where the covectors $d\xi$ and $d\eta$ are simultaneously zero only at the origin, it follows that F has no critical points in U other than p.

Step 3: Let H denote the closure of $F^{-1}(-\infty, c-\varepsilon] - M^{c-\varepsilon}$. The region $F^{-1}(-\infty, c-\varepsilon] = M^{c-\varepsilon} \cup H$ is a deformation retract of $M^{c+\varepsilon}$.
Consider the region $F^{-1}[c-\varepsilon, c+\varepsilon]$. By Step 1 together with the inequality $F \leq f$ we have $F^{-1}[c-\varepsilon, c+\varepsilon] \subseteq f^{-1}[c-\varepsilon, c+\varepsilon]$. Since this region is closed and is contained in a compact, it is compact. It can contain no critical points of F, except possibly p (by Step 2). But p is not contained in $F^{-1}[c-\varepsilon, c+\varepsilon]$, since $F(p) < c-\varepsilon$. Hence, $F^{-1}[c-\varepsilon, c+\varepsilon]$ contains no critical points. Then, applying the First Theorem of Morse 2.6 we have that $F^{-1}(-\infty, c-\varepsilon]$ is a deformation retract of $F^{-1}(-\infty, c+\varepsilon]$. Thus, by applying Step 1 we prove the result.

Step 4: The λ-cell e^λ is contained in H.
Indeed, let $q \in e^\lambda$. Since $0 = \xi(p) \leq \xi(q) \leq \varepsilon$ and $\frac{\partial F}{\partial \xi} < 0$ we have that $F(q) \leq F(p) < c-\varepsilon$, which implies that q is contained in the interior of $F^{-1}(-\infty, c-\varepsilon]$. On the other hand, since $\xi(q) \leq \varepsilon$ and $\eta(q) = 0$, we have $f(q) \geq c-\varepsilon$, which implies that q is not contained in the interior of $M^{c-\varepsilon}$. The result follows by applying

the set inclusion $\mathring{A} - \mathring{B} \subseteq \overline{A - B}$ with $A = F^{-1}(-\infty, c - \varepsilon]$ and $B = M^{c-\varepsilon}$.

Step 5: The region $M^{c-\varepsilon} \cup e^{\lambda}$ is a deformation retract of $M^{c-\varepsilon} \cup H$.
Let us define a function $r_t : M^{c-\varepsilon} \cup H \to M^{c-\varepsilon} \cup H$ which is the identity outside U and in U we define it distinguishing between the three regions indicated in the following Figure 2.4.

i) Region 1: Within this region $\xi \leq \varepsilon$. We define:

$$r_t(x_1, \ldots, x_m) = (x_1, \ldots, x_\lambda, tx_{\lambda+1}, \ldots, tx_m).$$

It is well defined (the image is in $F^{-1}(-\infty, c - \varepsilon]$) since r_1 is the identity and for $t < 1$ we have $\eta(r_t)$ decreases with t, while $\frac{\partial F}{\partial \eta} > 0$, which implies $F(r_t)$ decreases with t.

Note that r_0 maps the region into e^{λ}, as indicated in the Figure 2.4 by the vertical arrows.

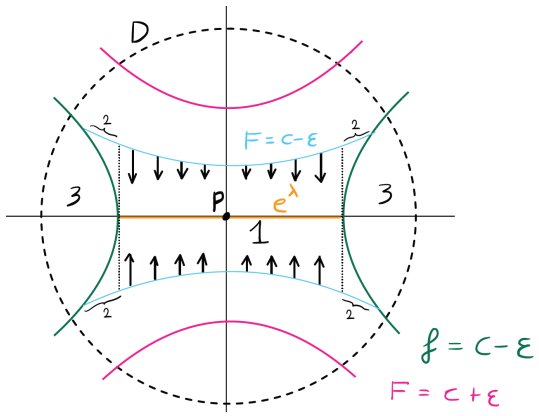

Figure 2.4: Schematic illustration using the same axes as in Figure 2.3. In this figure we also represent the sets D, e^{λ} and the hyperbola $f^{-1}(c - \varepsilon)$. We add the light blue and pink hyperbolas, which represent the hypersurfaces $F^{-1}(c - \varepsilon)$ and $F^{-1}(c + \varepsilon)$ respectively. The region enclosed by the three hyperbolas corresponds to the set $F^{-1}[c - \varepsilon, c + \varepsilon]$. The numbers 1, 2 and 3 indicate the different regions where we define r_t, and the arrows the retraction in the region 1. Regions 1 and 2 correspond to the set H, while region 3 corresponds to $M^{c-\varepsilon}$.

ii) Region 2: Within this region $\varepsilon \leq \xi \leq \eta + \varepsilon$. We define:

$$r_t(x_1, \ldots, x_m) = (x_1, \ldots, x_\lambda, s_t x_{\lambda+1}, \ldots, s_t x_m).$$

where $s_t \in [0, 1]$ is defined by $s_t := t + (1 - t)[\frac{\xi - \varepsilon}{\eta}]^{1/2}$.

Note that s_t is non-decreasing with t, since $\frac{ds_t}{dt} = 1 - [\frac{\xi - \varepsilon}{\eta}]^{1/2} \geq 0$. Then, as we did for region 1, since r_1 is the identity and for $t < 1$ we have $\eta(r_t)$ decreasing with

s_t (i.e with t), by the same argument r_t is well defined. In addition, this definition coincides with that of region 1 when $\xi = \varepsilon$.

Note that r_0 maps the region into $f^{-1}(c-\varepsilon)$, since $f(r_0) = c - \xi + s_0^2\eta = c - \varepsilon$. Also note that r_t is continuous in this region, since for $\lambda + 1 \leq i \leq m$ the functions $s_t x_i$ remain continuous when $\xi \to \varepsilon$, $\eta \to 0$ (note that $x_i \to 0$ since $\eta = \sum_{i=\lambda+1}^{m} x_i^2$ and $s_t x_i \to 0$).

iii) Region 3: Within this region $\eta + \varepsilon \leq \xi$, we define r_t simply as the identity. Note that this definition coincides with that of region 2 when $\xi = \eta + \varepsilon$.

Throughout $M^{c-\varepsilon} \cup H$, note that the defined r_t is a continuous function. The regions 1 and 2 together are the set H, while region 3 is in the set $M^{c-\varepsilon}$. Note that $r_0 : M^{c-\varepsilon} \cup H \to M^{c-\varepsilon} \cup e^\lambda$ is the identity in $M^{c-\varepsilon} \cup e^\lambda$ and is homotopic to the identity r_1. Therefore, r_0 defines a deformation retraction of $M^{c-\varepsilon} \cup H$ onto $M^{c-\varepsilon} \cup e^\lambda$.

□

Remark 2.12. (I) We can generalize the theorem supposing that there are k non-degenerate critical points $p_1, \ldots, p_k$ with indices $\lambda_1, \ldots, \lambda_k$ in $f^{-1}(c)$. Then a similar proof shows that $M^{c+\varepsilon}$ has the homotopy type of $M^{c-\varepsilon} \cup e^{\lambda_1} \cup \cdots \cup e^{\lambda_k}$.

(II) A simple modification of the proof shows that M^c is a deformation retract of $F^{-1}(-\infty, c]$, which is a deformation retract of $M^{c+\varepsilon}$. Then, M^c is a deformation retract of $M^{c+\varepsilon}$ and applying the theorem has the homotopy type of $M^{c-\varepsilon} \cup e^\lambda$.

2.4 Third Theorem of Morse

In this section, we introduce and prove what we refer to as the Third Theorem of Morse. This result characterizes the homotopy type of a compact manifold. Specifically, it states that for a given Morse function f, the compact manifold M on which f is defined has the homotopy type of a finite cell complex with one λ-cell for each critical point of f with index λ.

First, we introduce some definitions regarding cell attachments.

Definition 2.13. We define a **CW-complex** (or **cell complex**) X inductively in the following way:

1. Consider a discrete set X^0 of points, which are regarded as 0-cells.

2. For each $k \in \mathbb{N}$ we form the ***k*-skeleton** X^k from X^{k-1} by attaching k-cells e_i^k (for $i \in I$ where I is a non-empty set) to X^{k-1} via maps $g_i : \dot{e}_i^k \to X^{k-1}$. That is:
$$X^k := X^{k-1} \bigcup_{g_i} e_i^k = (X^{k-1} \bigsqcup_i e_i^k) / \sim_{g_i}$$
where we take the topological sum of all the k-cells with X^{k-1} and then we identify each $u \in \dot{e}_i^k$ with its image $g_i(u) \in X^{k-1}$.

We can set $X = X^k$ for some fixed k. In this case, X is said to be **finite-dimensional** of dimension k (the maximum dimension of cells of X). Also, we can continue the process indefinitely, setting $X = \bigcup_k X^k$ and considering in X the weak topology: a set $U \subseteq X$ is open (closed) if and only if $U \cap X^k$ is open (closed) in X^k for each $k \in \mathbb{N}$.
When the CW-complex X has finitely many cells, we say that it is **finite**. Note that finite CW-complexes are precisely the finite-dimensional CW-complexes whose skeletons each contain only finitely many cells.

Example 2.14. 1. The k-cells are CW-complexes constructed by attaching a k-cell to the empty set.
2. The boundaries $\dot{e}^k = S^{k-1}$ are CW-complexes constructed by attaching a $(k-1)$-cell e^{k-1} to a 0-cell e^0 (which is just a point). This is immediate from the definition, since the attaching map $g : \dot{e}^{k-1} \to e^0$ just identifies $\dot{e}^{k-1}$ with the point e^0, therefore the CW-complex is $e^{k-1}/\dot{e}^{k-1} = D^{k-1}/S^{k-2} = S^{k-1} = \dot{e}^k$.

Definition 2.15. A **subcomplex** of a cell complex X is a closed subspace $A \subseteq X$ that inherits a cell complex structure from X, consisting of the union of some of its cells. A pair (X, A) consisting of a cell complex X and a subcomplex A is called a **CW pair**.

Example 2.16. Each k-skeleton X^k or k-cell e^k of a cell complex X is a subcomplex.

Definition 2.17. Let X and Y be CW-complexes. We say that a continuous map $f : X \to Y$ is a **cellular map** if the image of the k-skeletons of X, namely X^k, lie in the k-skeletons of Y:
$$f(X^k) \subseteq Y^k$$
for each $k \in \mathbb{N}$ such that both X and Y contain k-cells.

Cellular maps are those that respect the cellular structures of the CW-complexes. In general, a map between CW-complexes is not cellular, but we have the following result:

Theorem 2.18. (Cellular Approximation Theorem) Let X and Y be CW-complexes. Every continuous map $f : X \to Y$ is homotopic to a cellular map.

Proof. We refer the reader to proof of Theorem 4.8 in [4]. □

We now present two lemmas concerning the extension of homotopy equivalences under cellular attachments, which will be used in the proof of the Third Theorem of Morse. A formal proof of these lemmas is provided in Appendix A.

Lemma 2.19. Let X be a topological space, $k \in \mathbb{N}$. Suppose $g_0, g_1 : \dot{e}^k \to X$ are homotopic attaching maps. Then, the identity map of X extends to a homotopy equivalence

$$h : X \cup_{g_0} e^k \to X \cup_{g_1} e^k.$$

Lemma 2.20. Let X be a topological space, $k \in \mathbb{N}$. Let $\varphi : \dot{e}^k \to X$ be an attaching map. Then, any homotopy equivalence $f : X \to Y$ extends to a homotopy equivalence

$$F : X \cup_{\varphi} e^k \to Y \cup_{f \circ \varphi} e^k.$$

We are now ready to prove the Third Theorem of Morse.

Theorem 2.21. (Third Theorem of Morse) Let M be a compact manifold and $f \in C^\infty(M)$ be a Morse function. Then, M has the homotopy type of a finite CW-complex with one λ-cell attached for each critical point of index λ.

Proof. Let $c_1 < c_2 < c_3 < \cdots < c_n$ be the critical values of f, which are finitely many by Corollary 2.4. Suppose there exists $a \in \mathbb{R}$ satisfying $a \notin \{c_i\}_{1 \leq i \leq n}$ such that M^a has the homotopy type of a CW-complex X, with a homotopy equivalence $h_1 : M^a \to X$. Let $c := \min_{1 \leq i \leq n} \{c_i : c_i > a\}$. By applying the Second Theorem of Morse 2.11 together with Remark 2.12, we have the following result:

For all sufficiently small $\varepsilon > 0$ such that the compact $f^{-1}[c - \varepsilon, c + \varepsilon]$ contains only the critical points of critical value c, namely $p_1, \ldots, p_{j(c)}$ for some $j(c) \in \mathbb{N}$ and with indices $\lambda_1, \ldots, \lambda_{j(c)}$, there exists a homotopy equivalence

$$M^{c-\varepsilon} \cup_{\varphi_1} e^{\lambda_1} \cdots \cup_{\varphi_{j(c)}} e^{\lambda_{j(c)}} \to M^{c+\varepsilon} \tag{2.3}$$

where $\varphi_i : \dot{e}^{\lambda_i} \to M^{c-\varepsilon}$ are attaching maps for $1 \leq i \leq j(c)$. Furthermore, by applying the First Theorem of Morse 2.6 there exists a homotopy equivalence $h_2 : M^{c-\varepsilon} \to M^a$.

Let us consider, for $1 \leq i \leq j(c)$, the attaching maps $\phi_i := h_1 \circ h_2 \circ \varphi_i : \dot{e}^{\lambda_i} \to X$. By applying Lemma 2.20, the homotopy equivalence $h_1 \circ h_2 : M^{c-\varepsilon} \to X$ extends to a homotopy equivalence

$$M^{c-\varepsilon} \cup_{\varphi_1} e^{\lambda_1} \cdots \cup_{\varphi_{j(c)}} e^{\lambda_{j(c)}} \to X \cup_{\phi_1} e^{\lambda_1} \cdots \cup_{\phi_{j(c)}} e^{\lambda_{j(c)}} \tag{2.4}$$

Combining the homotopy equivalences (2.3) and (2.4), we have a homotopy equivalence:

$$X \cup_{\phi_1} e^{\lambda_1} \cdots \cup_{\phi_{j(c)}} e^{\lambda_{j(c)}} \to M^{c+\varepsilon}. \tag{2.5}$$

Although the ϕ_i map the CW-complexes $\dot{e}_i^{\lambda}$ to the CW-complex X, the attachment $X \cup_{\phi_1} e^{\lambda_1} \cdots \cup_{\phi_{j(c)}} e^{\lambda_{j(c)}}$ is not a CW-complex in general, since the attachments are not necessarily made within the appropriate skeletons of X. However, by the Cellular Approximation Theorem 2.18, there exist cellular attaching maps to the appropriate skeletons of X:

$$\psi_i : \dot{e}^{\lambda_i} \to X^{\lambda_i - 1},$$

for $1 \leq i \leq j(c)$, which are homotopic to the ϕ_i. Then, by applying Lemma 2.19 we have a homotopy equivalence

$$X \cup_{\phi_1} e^{\lambda_1} \cdots \cup_{\phi_{j(c)}} e^{\lambda_{j(c)}} \to X \cup_{\psi_1} e^{\lambda_1} \cdots \cup_{\psi_{j(c)}} e^{\lambda_{j(c)}} \tag{2.6}$$

where the attachment with the maps ψ_i is a CW-complex. Hence, by combining (2.5) and (2.6), we have that $M^{c+\varepsilon}$ has the homotopy type of the CW-complex $X \cup_{\psi_1} e^{\lambda_1} \cdots \cup_{\psi_{j(c)}} e^{\lambda_{j(c)}}$.

Now, let us apply this result to the arbitrary compact manifold M. First, observe that since M is compact, by Weierstrass' Theorem, the function f attains a maximum and a minimum value. Since these are critical values, we have that c_1 and c_n are the minimum and maximum values of f, respectively. Then, we have that $M^a = \emptyset$ for $a < c_1$ and $M^a = M$ for $a > c_n$.

By applying the deduced result to $M^a = \emptyset$ for $a < c_1$, we have that $M^{c_1+\varepsilon}$ has the homotopy type of the CW-complex $e^{\lambda_1^1} \cup \cdots \cup e^{\lambda_{j(c_1)}^1}$ for a sufficiently small ε satisfying $0 < \varepsilon < c_2 - c_1$, where λ_i^1 denote the indices of the critical points with critical value c_1. By applying the First Theorem of Morse 2.6 between $c_1 + \varepsilon$ and all $a \in (c_1, c_2)$, we have that M^a has the homotopy type of $M_1 := e^{\lambda_1^1} \cup \cdots \cup e^{\lambda_{j(c_1)}^1}$ for all $c_1 \leq a < c_2$ (the inequality $\leq$ comes from Remark 2.12.)

We can apply this reasoning inductively to get that for $2 \leq i \leq n-1$, M^a has the homotopy type of the CW-complex $M_i := M_{i-1} \cup e^{\lambda_1^i} \cup \cdots \cup e^{\lambda_{j(c_i)}^i}$ for $c_i \leq a < c_{i+1}$.

Finally, by applying the First Theorem of Morse 2.6 between values $a > c_n$, we have that $M^a = M$ has the homotopy type of the CW-complex $M_n := M_{n-1} \cup e^{\lambda_1^n} \cup \cdots \cup e^{\lambda_{j(c_n)}^n}$ for $a \geq c_n$. By construction of M_n, this proves the result. □

Remark 2.22. (I) We have also proved that each M^a has the homotopy type of a finite CW-complex, with one cell of dimension λ for each critical point of index λ in M^a. This is true even if a is a critical value, according to Remark 2.12.

(II) The theorem also holds without assuming that M is compact, provided that M^a is compact for each $a \in \mathbb{R}$. If M is not compact, but all critical points lie in one of the compact sets M^a, then a proof similar to the First Theorem of Morse 2.6 shows that M^a is a deformation retract of M, which implies M has the stated homotopy type. If this is not the case, we refer the reader to the proof of Theorem 3.5 in [1].

(III) A given Morse function on a compact manifold determines a particular CW decomposition, where each critical point of index λ corresponds to a λ-cell. However, different Morse functions on the same manifold can have different numbers and indices of critical points, and thus induce different CW structures. This does not contradict the theorem, since a compact manifold can have the homotopy type of multiple CW-complexes. Nevertheless, as we will see in the next chapter, homotopy invariants of the manifold impose lower bounds on the number of critical points of each index for any Morse function. This implies that the minimal number of cells in each dimension of a CW-complex homotopy equivalent to M is constrained by the topology of the manifold.

2.5 Example

In this section, illustrate the use of the results deduced in this chapter in the example of the height function on a torus.

The height function is particularly useful for gaining insight into the results we studied. This is the function we have used in some previous illustrations. Formally, it is defined as:

Definition 2.23. Let M be a compact smooth manifold embedded in the Euclidean space $(\mathbb{R}^n, \langle \cdot, \cdot \rangle)$. Given a unit vector $v \in \mathbb{R}^n$, the **height function** associated to v is the map:

$$h_v : M \to \mathbb{R}, \quad h_v(p) = \langle p, v \rangle.$$

Example 2.24. For a compact regular surface embedded in $\mathbb{R}^3$ with respect to some axes, we have the usual height function $h(x, y, z) = z$ (which corresponds to $v = (0, 0, 1)$).

The geometric interpretation of h_v is that it measures the vertical coordinate of a point $p \in M$ with respect to the direction given by v. Equivalently, it corresponds to the projection of M onto the line spanned by v. The height function is smooth, and on a compact manifold it attains both a maximum and a minimum value.

An example is illustrated in Figure 2.5 for the case of the torus, which is a smooth manifold of dimension 2. For this manifold, the critical points of a height function h are all non-degenerate. Thus, h is a Morse function, and we can apply the results we have deduced.

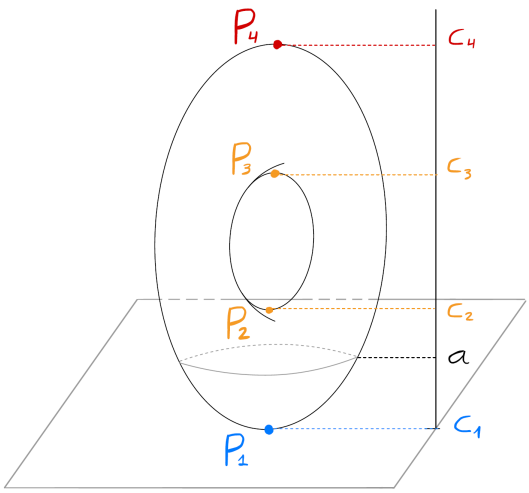

Figure 2.5: Illustration of a torus and its vertical height function h. The four coloured points p_i are the non-degenerate critical points of h on the torus, with corresponding critical values $c_i = h(p_i)$. The height a is an arbitrary value between c_1 and c_2, and the points of the torus with height less than or equal to a form the subset M^a.

Let λ_i be the index of a the critical point p_i. According to Morse's Lemma, note that:

1. At the minimum, located at the critical point p_1, the function h can be locally parametrized as $h = c_1 + x^2 + y^2$ for some local coordinates x, y. Thus, $\lambda_1 = 0$.
2. At the saddle points, located at p_2 and p_3, the function h can be locally parametrized as $h = c_{2,3} - x^2 + y^2$, and $\lambda_2 = \lambda_3 = 1$.
3. At the maximum, located at p_4, the function h can be locally parametrized as $h = c_4 - x^2 - y^2$, and $\lambda_4 = 2$.

Observe that the subsets M^a, for heights a in a same interval of critical values (c_i, c_{i+1}), are homeomorphic to one another, according to the First Theorem of Morse. This is not the case when we consider subsets M^a for values of a lying in distinct intervals (c_i, c_{i+1}). These cases can be interpreted in terms of the Second Theorem of Morse, as shown in the following figure:

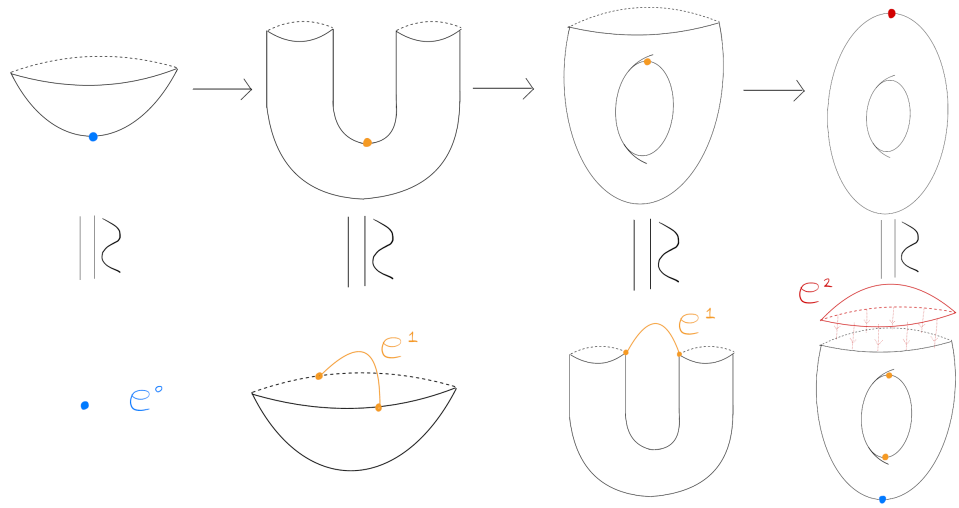

Figure 2.6: Illustration of the different subsets M^a, depending on which interval (c_i, c_{i+1}) the height a lies in, and their corresponding homotopy equivalents, in the case of the height function defined on a torus. The subsets in the upper part represent, from left to right, the M^a with $a \in (c_i, c_{i+1})$ for $i = 1, 2, 3$, and for $a \geq c_4$. In the lower part, for each of these subsets, their homotopy equivalents are shown, constructed by attaching cells as described by the Second Theorem of Morse. The vertical symbol $\cong$ denotes the homotopy equivalence relation, and the e^k denote the attached k-cells.

1. For $a < c_1$, the subset M^a is empty.
2. For $c_1 \leq a < c_2$, the subsets M^a are homeomorphic to a 2-cell. Observe that a 2-cell has the same homotopy type as a 0-cell, in accordance with the Second Theorem of Morse when a passes through the critical point c_1 of index $\lambda_1 = 0$.
3. For $c_2 \leq a < c_3$, the subsets M^a are homeomorphic to a cylinder. Observe that a cylinder has the same homotopy type as a 2-cell with a 1-cell attached, in accordance with $\lambda_2 = 1$
4. For $c_3 \leq a < c_4$, the subsets M^a are homeomorphic to a compact manifold of genus 1 with a circle as boundary. Observe that this manifold has the same homotopy type as a cylinder with a 1-cell attached, in accordance with $\lambda_3 = 1$.
5. For $a \geq c_4$, the subsets M^a coincide with M. Observe that the torus M has

the same homotopy type as the previous manifold of genus 1 with a 2-cell attached, in accordance with $\lambda_4 = 2$.

Finally, observe that from the indices of the critical points of the height function, and by applying the Third Theorem of Morse, we know that the torus has the homotopy type of a CW-complex with one 0-cell, two 1-cells, and one 2-cell.

This example illustrates how Morse theory can provide insight into the homotopy structure of a compact manifold and its subspaces. Conversely, topological considerations also provide information about the functions. For instance, Weierstrass' Theorem guarantees that such functions have at least two critical points: a maximum and a minimum. In the next chapter, we will explore this idea further. Specifically, we will use Morse theory to learn about Morse functions from the homotopy invariants of the manifold.

Chapter 3

Morse Inequalities

In this chapter, we introduce and demonstrate the Morse inequalities. For a Morse function defined on a compact manifold, Morse inequalities relate the homological structure of the manifold to the critical points of the function. Specifically, they are expressions that relate the homotopy invariants of the manifold–the Betti numbers and the Euler characteristic–to the number of critical points of the Morse function. We begin the chapter by presenting these topological invariants, and then we prove the aforementioned results.

3.1 Homology with Coefficient Groups and Homotopy Invariants

In this section, we study the structure of finitely generated homology groups for arbitrary coefficient groups. We focus on the case when the coefficient group is the additive group of a field F, in which case the homology groups become F-vector spaces. We introduce the Betti numbers and the Euler characteristic, which are homotopy invariants, and prove that they are subadditive and additive, respectively.

Let us consider a pair (A, B), where B is a subspace of the topological space A. We denote by $H_n(A, B; \mathbb{Z})$ the relative integral singular homology group of the pair (A, B). The singular relative homology groups are homotopy invariants, meaning that homotopy equivalent spaces have isomorphic homology groups.

Throughout this section, let us suppose that the groups $H_n(A, B; \mathbb{Z})$ are finitely generated abelian groups. Then, by the Fundamental Theorem of Finitely Generated Abelian Groups, they can be decomposed as:

$$H_n(A,B;\mathbb{Z}) \cong \mathbb{Z}^r \oplus \frac{\mathbb{Z}}{d_1\mathbb{Z}} \oplus \frac{\mathbb{Z}}{d_2\mathbb{Z}} \oplus \cdots \oplus \frac{\mathbb{Z}}{d_k\mathbb{Z}}, \tag{3.1}$$

where $\cong$ denotes isomorphism, $r \in \mathbb{N} \cup \{0\}$ is named the rank, and the integers $d_i > 1$, which satisfy $d_i \mid d_{i+1}$, are the invariant factors. Naturally, the rank and the invariant factors depend on n, but we do not make this dependence explicit.

Definition 3.1. For (A,B), where B is a subspace of the topological space A, we define the nth **Betti number** (with coefficients in $\mathbb{Z}$) of (A,B) as

$$b_n(A,B;\mathbb{Z}) := \operatorname{rank}(H_n(A,B;\mathbb{Z})),$$

provided that $H_n(A,B;\mathbb{Z})$ is finitely generated.

When the homology groups are computed for an arbitrary coefficient group G, the decomposition of $H_n(A,B;G)$ is given by the Universal Coefficients Theorem for Homology[1]

$$H_n(A,B;G) \cong H_n(A,B;\mathbb{Z}) \otimes G \oplus \operatorname{Tor}_1^{\mathbb{Z}}(H_{n-1}(A,B;\mathbb{Z}),G), \tag{3.2}$$

where $\operatorname{Tor}_1^{\mathbb{Z}}(J,K)$ is a torsion abelian group, for abelian groups J and K, endowed with the natural structure of $\mathbb{Z}$-modules[2].

Let us focus on the two terms that are summed up in (3.2).

First, if we tensor product the decomposition (3.1) by $\otimes G$, we end up with

$$H_n(A,B;\mathbb{Z}) \otimes G \cong G^r \oplus \frac{G}{d_1G} \oplus \frac{G}{d_2G} \oplus \cdots \oplus \frac{G}{d_kG}, \tag{3.3}$$

where we have distributed the tensor product with respect to the direct sum, and applied that $\mathbb{Z} \otimes G \cong G$ and $\frac{\mathbb{Z}}{d\mathbb{Z}} \otimes G \cong \frac{G}{dG}$ for d a natural number [3]. Note that the torsion-free part of the decomposition (3.2) is G^r, thus the rank of $H_n(A,B;G)$ coincides with the rank of $H_n(A,B;\mathbb{Z})$.

Let us now analyse the other summand in (3.2). For a fixed abelian group K, it is satisfied that $\operatorname{Tor}_1^{\mathbb{Z}}(J,K) = 0$ for every abelian group J if and only if K is torsion-free[4]. This condition implies that independently of $H_{n-1}(A,B;\mathbb{Z})$, if G is

[1] A proof of this theorem can be found in Chapter 9, section *Universal Coefficients*, of [7].
[2] A formal definition of the Tor term can be found in Chapters 2 and 3, of [8].
[3] These properties are proved in Chapter 9, sections *Tensor Product* and *Universal Coefficients*, of [7].
[4] This condition can be found in §3.2, of [8].

torsion-free, the term $\mathrm{Tor}_1^{\mathbb{Z}}(H_{n-1}(A,B;\mathbb{Z}),G)$ vanishes.

An interesting case occurs when the coefficient group is a field F (more precisely, the additive group of a field). In this case, since a field has only trivial ideals, the torsion terms in (3.3) vanish. The Tor term in (3.2) does not vanish in general, but has a vector space structure over the field F[5]. Then, since F^r is a F-vector space and the direct sum of vector spaces is a vector space, by (3.2) the homology groups $H_n(A,B;F)$ have F-vector space structure. Then, when taking a field F as the coefficient group, we talk about singular **homology vector spaces** $H_n(A,B;F)$.

Note that if F is torsion-free, then the homology vector spaces are torsion-free, and it is satisfied that $H_n(A,B;F) \cong F^r$. This is true since in this case, the Tor term in (3.2) vanishes. An equivalent condition for F to be torsion-free (as a $\mathbb{Z}$-module) is to have zero characteristic. This is an immediate consequence of the definitions, since for every non-zero integer n and every non-zero element $a \in F$, the torsion-free condition $n \cdot a \neq 0$ is equivalent to the zero characteristic condition $n \cdot 1 \neq 0$, by multiplying and dividing by a. Therefore, for zero characteristic fields such as $\mathbb{Q}, \mathbb{R}$ or $\mathbb{C}$, the rank r completely characterizes the homology vector spaces.

The preceding discussion suggests a natural generalization of the Betti number, which will be considered when presenting the Morse inequalities.

Definition 3.2. For (A,B), where B is a subspace of the topological space A, we define the nth **Betti number** of (A,B) **with coefficients in a field** F as

$$b_n(A,B;F) := \dim_F(H_n(A,B;F)).$$

Observe that for a zero characteristic field, $b_n(A,B;F)$ coincides with the previous Betti number definition. If F has non-zero characteristic (for instance, the group $\frac{\mathbb{Z}}{p\mathbb{Z}}$ where the prime p is the characteristic), then $b_n(A,B;F)$ does not coincide with $b_n(A,B;\mathbb{Z})$. In this case, the Betti number with coefficients in the field may also capture torsion effects from lower degree homology vector spaces from the Tor term in the decomposition (3.2).

Another important concept for establishing the Morse inequalities is the following.

[5]This condition holds since $\mathrm{Tor}_1^R(J,K)$ is a S-module when K is a $R-S$ bimodule, where R and S are rings. In particular, when $K = F$ is a field, it has compatible left $\mathbb{Z}$-module and right F-module structures, given by the natural left action of $\mathbb{Z}$ on an abelian group and the right multiplication of the field, respectively. Therefore, F is a $\mathbb{Z}-F$ bimodule, and $\mathrm{Tor}_1^{\mathbb{Z}}(J,F)$ is a F-vector space. More details can be found in § 2.6, of [8].

Definition 3.3. For (A, B), where B is a subspace of the topological space A, we define the **Euler characteristic** of (A, B) **with coefficients in a field** F as

$$\chi(A, B; F) := \sum_{n=0}^{\infty} (-1)^n b_n(A, B; F).$$

This definition is a generalization of the definition of the Euler characteristic for a finite CW-complex X:

$$\chi(X; \mathbb{Z}) := \sum_{n=0}^{\infty} (-1)^n c_n(X)$$

where $c_n(X)$ is the number of n-cells of X. Indeed, when F is a zero characteristic field, the Betti numbers $b_n(X; F)$ coincide with the ranks of the integral singular homology groups, and the alternating sum of ranks equals the alternating sum of the numbers of cells[6].

Observe that, since the singular homology groups are homotopy invariant, both the Betti numbers and the Euler characteristic are homotopy invariants by definition.

From now on, we consider homology vector spaces with coefficients in a fixed field F. We do not explicit the field F and simplify the notation: $H_n(A, B; F) = H_n(A, B)$, $b_n(A, B; F) = b_n(A, B)$ and $\chi(A, B; F) = \chi(A, B)$.

An important property of the Betti numbers and the Euler characteristic is that they are subadditive and additive, respectively.

Definition 3.4. Let S be a function that sends pairs of spaces (A, B) to integers $S(A, B) \in \mathbb{Z}$. We say that S is **subadditive** if for spaces $C \subseteq B \subseteq A$ we have

$$S(A, C) \leq S(A, B) + S(B, C).$$

If the equality holds, we say that S is **additive**.

For a chain of spaces $A_0 \subseteq \cdots \subseteq A_k$, subadditivity of S implies

$$S(A_k, A_0) \leq \sum_{i=1}^{k} S(A_i, A_{i-1}),$$

which follows immediately from the definition by induction on k. If S is additive, the equality holds. If we let $S(A, \emptyset) = S(A)$, where $S(\cdot)$ is another function that sends spaces A to integers, then by taking $A_0 = \emptyset$, we have the inequality $S(A_k) \leq \sum_{i=1}^{k} S(A_i, A_{i-1})$, with equality when S is additive.

[6] A proof of this last statement can be found in §2.2, of [4].

Proposition 3.5. Let B be a subspace of the space A. Then:
a) The nth Betti number $b_n(A,B)$ is subadditive.
b) The Euler characteristic $\chi(A,B)$ is additive.
c) The function $s_n(A,B) := b_n(A,B) - b_{n-1}(A,B) + b_{n-2}(A,B) - \cdots \pm b_0(A,B)$ is subadditive.

Proof. Let us consider the exact sequence for a triple of spaces (A,B,C), where $C \subseteq B \subseteq A$:

$$\ldots \xrightarrow{\partial_{n+1}} H_n(B,C) \xrightarrow{\varepsilon_n} H_n(A,C) \xrightarrow{\sigma_n} H_n(A,B) \xrightarrow{\partial_n} \ldots \xrightarrow{\sigma_0} H_0(A,B) \xrightarrow{\partial_0} 0$$

for linear maps $\partial_n, \varepsilon_n$ and σ_n.

(a) Since the sequence is exact, by definition $\mathrm{Ker}(\sigma_n) \cong \mathrm{Im}(\varepsilon_n)$. Then, by applying the First Isomorphism Theorem to σ_n, we have $\mathrm{Im}(\sigma_n) \cong H_n(A,C)/\mathrm{Im}(\varepsilon_n)$, and the dimensional identity

$$b_n(A,C) = \dim(H_n(A,C)) = \dim(\mathrm{Im}(\sigma_n)) + \dim(\mathrm{Im}(\varepsilon_n)). \tag{3.4}$$

The first term on the right-hand side is bounded by $b_n(A,B)$, since $\mathrm{Im}(\sigma_n) \subseteq H_n(A,B)$. The second term is bounded by $b_n(B,C)$, since $\mathrm{Im}(\varepsilon_n) \cong H_n(B,C)/\mathrm{Ker}(\varepsilon_n)$. This proves subadditivity of the nth Betti number.

(b) Analogous dimensional identities to (3.4) hold from applying the exact condition and the First Isomorphism Theorem to ε_n and ∂_n:

$$b_n(B,C) = \dim(\mathrm{Im}(\partial_{n+1})) + \dim(\mathrm{Im}(\varepsilon_n)), \tag{3.5}$$

$$b_n(A,B) = \dim(\mathrm{Im}(\sigma_n)) + \dim(\mathrm{Im}(\partial_n)). \tag{3.6}$$

If we express the Euler characteristics of the three pairs, by the previous identities (3.4), (3.5) and (3.6), we obtain:

$$\chi(A,C) = \sum_n (-1)^n \dim(\mathrm{Im}(\sigma_n)) + \sum_n (-1)^n \dim(\mathrm{Im}(\varepsilon_n)), \tag{3.7}$$

$$\chi(B,C) = \sum_n (-1)^n \dim(\mathrm{Im}(\partial_{n+1})) + \sum_n (-1)^n \dim(\mathrm{Im}(\varepsilon_n)), \tag{3.8}$$

$$\chi(A,B) = \sum_n (-1)^n \dim(\mathrm{Im}(\sigma_n)) + \sum_n (-1)^n \dim(\mathrm{Im}(\partial_n)). \tag{3.9}$$

Note that, by changing the index n to $n-1$ in the first term on the right-hand side of (3.8), this transforms into $-\sum_n (-1)^n \dim(\mathrm{Im}(\partial_n))$. Then, if we sum (3.8) and (3.9) and compare with (3.7), we obtain the additive condition for the Euler characteristic.

(c) By successively substituting terms from the expressions (3.4), (3.5) and (3.6), we obtain:

$$\begin{aligned} 0 \leq \dim(\mathrm{Im}(\partial_{n+1})) &= b_n(B,C) - \dim(\mathrm{Im}(\varepsilon_n)) \\ &= b_n(B,C) - b_n(A,C) + \dim(\mathrm{Im}(\sigma_n)) \\ &= b_n(B,C) - b_n(A,C) + b_n(A,B) - \dim(\mathrm{Im}(\partial_n)). \end{aligned}$$

By continuing the substitution process as n decreases, we directly obtain:

$$0 \leq s_n(B,C) - s_n(A,C) + s_n(A,B)$$

which proves the subadditivity of s_n.

□

3.2 Weak and Strong Morse Inequalities

In this section, we introduce and demonstrate the Morse inequalities, which relate the Betti numbers and the Euler characteristic of a compact manifold M to the number of critical points of a Morse function f defined on the manifold. Specifically, we show that the λ-Betti number is a lower bound for the number of critical points of f with index λ, and that modulo 2 the total number of critical points equals the Euler characteristic of the manifold.

First, we discuss why we can apply the results deduced in the previous section to the case of a compact manifold.

When discussing the singular homology groups of a compact manifold M, by homotopy invariance, one can instead consider the singular homology groups of a finite CW-complex homotopy equivalent to M (which exists as shown in Theorem 2.21). The same reasoning applies to subcomplexes, such as the subcomplex homotopy equivalent to M^a (see Remark 2.22). Moreover, we identify the singular homology groups of a CW complex with its isomorphic cellular homology groups. This identification is useful because it allows us to apply properties of cellular homology, which simplify the computation of the singular homology groups.

A relevant first property is that for a CW-complex, the nth integral homology group has a number of generators less or equal than the number of n-cells in the complex. A second one is that for a subcomplex Y of a CW-complex X, there is an isomorphism $H_n(X,Y;\mathbb{Z}) \cong \tilde{H}_n(X/Y;\mathbb{Z})$, where $\tilde{H}_n(X/Y;\mathbb{Z}) \cong H_n(X/Y;\mathbb{Z})$

except for $n = 0$, in which case we have $\tilde{H}_0(X/Y;\mathbb{Z}) \oplus \mathbb{Z} \cong H_0(X/Y;\mathbb{Z})$.

For a finite CW-complex X and any subcomplex Y, the quotient X/Y is also finite, and therefore the integral homology groups $H_n(X;\mathbb{Z})$ and $H_n(X/Y;\mathbb{Z})$ are finitely generated by the first property. In addition, by the second property, this implies that the relative integral homology groups $H_n(X,Y;\mathbb{Z})$ are also finitely generated.

Since a compact manifold M is homotopy equivalent to a finite CW-complex, this implies that the relative integral homology groups of M and its subspaces are finitely generated. Therefore, we can apply the results deduced in the previous section and focus on relative homology vector spaces with coefficients in a fixed field, which we do not make explicit in the notation.

We are now ready to prove the Morse inequalities:

Theorem 3.6. Let M be a compact manifold and $f \in C^\infty(M)$ be a Morse function. Let $c_\lambda(f)$ be the number of critical points of f with index λ, $b_\lambda(M)$ be the Betti numbers, and $\chi(M)$ be the Euler characteristic. Then:

1. **Weak Morse inequalities**:

$$b_\lambda(M) \leq c_\lambda(f), \quad \text{and} \quad \chi(M) = \sum_{\lambda=0}^{\infty} (-1)^\lambda c_\lambda(f).$$

2. **Strong Morse inequalities**:

$$b_\lambda(M) - b_{\lambda-1}(M) + \cdots \pm b_0(M) \leq c_\lambda(f) - c_{\lambda-1}(f) + \cdots \pm c_0(f).$$

Proof. Let X be the CW-complex homotopy equivalent to M provided by the Third Theorem of Morse 2.21 for the Morse function f. By homotopy invariance, the Betti numbers and Euler characteristic of M and X coincide, so we can work with X instead of M. For a $\lambda \in \mathbb{N} \cup \{0\}$, the number of λ cells in X is equal to the number of critical points of f of index λ: $c_\lambda(f)$. Let $\lambda_{max} \leq m$ be the largest index of the critical points of f. Then, we have a chain of the k-skeletons of X:

$$\emptyset = X^{-1} \subseteq X^0 \subseteq \cdots \subseteq X^{\lambda_{max}} = X.$$

By Proposition 3.5, we have:

$$b_\lambda(X) \leq \sum_{i=0}^{\lambda_{max}} b_\lambda(X^i, X^{i-1}), \tag{3.10}$$

$$\chi(X) = \sum_{i=0}^{\lambda_{max}} \chi(X^i, X^{i-1}), \tag{3.11}$$

$$s_\lambda(X) \leq \sum_{i=0}^{\lambda_{max}} s_\lambda(X^i, X^{i-1}). \tag{3.12}$$

Now, note that the relative Betti numbers for the skeletons of X are[7]

$$b_\lambda(X^i, X^{i-1}) = \begin{cases} c_\lambda(f) & \text{for } \lambda = i, \\ 0 & \text{for } \lambda \neq i. \end{cases}$$

Hence, the sum on the right-hand side of the inequality (3.10) is just $c_\lambda(f)$, which proves the first weak Morse inequality for arbitrary λ. Also, since

$$\chi(X^i, X^{i-1}) = \sum_{k=0}^{\infty} (-1)^k b_k(X^i, X^{i-1}) = (-1)^i c_i(f),$$

comparing with (3.11) we obtain the second weak Morse inequality (note that $c_\lambda(f) = 0$ for $\lambda > \lambda_{max}$). Finally, since

$$s_\lambda(X^i, X^{i-1}) = b_\lambda(X^i, X^{i-1}) - b_{\lambda-1}(X^i, X^{i-1}) + \cdots \pm b_0(X^i, X^{i-1})$$

the sum on the right-hand side of the inequality (3.12) is equal to

$$\begin{aligned} &\sum_{i=0}^{\lambda_{\max}} b_\lambda(X^i, X^{i-1}) - \sum_{i=0}^{\lambda_{\max}} b_{\lambda-1}(X^i, X^{i-1}) + \cdots \pm \sum_{i=0}^{\lambda_{\max}} b_0(X^i, X^{i-1}) = \\ &= \quad c_\lambda(f) - c_{\lambda-1}(f) + \cdots \pm c_0(f) \end{aligned}$$

which proves the strong Morse inequality for arbitrary λ. □

Remark 3.7. The weak Morse inequalities can be obtained directly from the strong Morse inequalities. Indeed, the first weak Morse inequality for an arbitrary λ is obtained by adding the strong inequalities for λ and $\lambda - 1$. Additionally, if we compare the strong inequalities for $\lambda = m + 1$ and $\lambda = m$ we obtain the second weak Morse inequality.

[7]Here we use that $H_\lambda(X^i, X^{i-1})$ is zero for $\lambda \neq i$ and is a vector space for $\lambda = i$, with a basis in one-to-one correspondence with the i-cells of X. This follows from the isomorphism $H_\lambda(X^i, X^{i-1}) \cong \tilde{H}_\lambda(X^i/X^{i-1})$, where X^i/X^{i-1} is a wedge sum of i-spheres. The result then uses the facts that the reduced homology of a wedge sum is the direct sum of the reduced homologies of its summands, and that the λth reduced homology vector space of an i-sphere is one-dimensional for $\lambda = i$ and zero otherwise.

For any Morse function f defined on a compact manifold M, weak Morse inequalities for the Betti numbers give lower bounds on the number of critical points of f of each index. Also, the Euler characteristic determines the parity of the total number of critical points. This last statement follows from observing that if $c(f) = \sum_\lambda c_\lambda(f)$ is the total number of critical points, then $c(f) - \chi(M) = \sum_\lambda [1 - (-1)^\lambda] c_\lambda(f) \equiv 0 \bmod 2$.

Furthermore, Morse inequalites are useful to gain insight about the homology groups of a given compact manifold, when the critical points of a Morse function are known. Let us give the following corollary as an example of this:

Corollary 3.8. With the same notation as in the previous theorem, if $c_{\lambda+1}(f) = c_{\lambda-1}(f) = 0$, then $b_{\lambda+1}(M) = b_{\lambda-1}(M) = 0$ and $b_\lambda(M) = c_\lambda(f)$.

Proof. Suppose $c_{\lambda+1}(f) = c_{\lambda-1}(f) = 0$. The first weak Morse inequalities directly give $b_{\lambda+1}(M) = b_{\lambda-1}(M) = 0$. Then, if we compare the strong Morse inequalities for λ and $\lambda + 1$, we obtain that the strong Morse inequality for λ is an equality. If we do the same for $\lambda - 2$ and $\lambda - 1$, we obtain that the inequality is an equality also for the case $\lambda - 2$. By subtracting this equality from the previous one, we obtain the result. □

An equivalent formulation of the Morse inequalities can be expressed in polynomial form. To this end, we define two polynomials:

Definition 3.9. Let M be a compact manifold and $f \in C^\infty(M)$ be a Morse function. Let $b_\lambda = b_\lambda(M)$ be the λ Betti number of M and $m_\lambda = c_\lambda(f)$ be the number of critical points of f with index λ. We define:

1. The **Morse Polynomial** $m(f,t) := \sum_\lambda m_\lambda t^\lambda$.
2. The **Poincaré Polynomial** $b(M,t) := \sum_\lambda b_\lambda t^\lambda$.

Corollary 3.10. (Polynomial Morse inequalities) Strong Morse inequalities can be expressed as

$$m(f,t) - b(M,t) = (1+t)q(t)$$

where $q(t) = \sum_\lambda q_\lambda t^\lambda$ is a polynomial with $t, q_\lambda \geq 0$.

Proof. Let us express the strong Morse inequality as $\sum_k (-1)^k b_{\lambda-k} \leq \sum_k (-1)^k m_{\lambda-k}$. Equivalently, there exists a non-negative integer q_λ such that $\sum_k (-1)^k (m_{\lambda-k} - b_{\lambda-k}) = q_\lambda$. Multiplying both sides by t^λ and summing over all λ, we obtain $\sum_\lambda \sum_k (-1)^k (m_{\lambda-k} - b_{\lambda-k}) t^\lambda = \sum_\lambda q_\lambda t^\lambda$. Now, performing the change of variable

$\lambda - k \to \lambda$ in the double sum on the left-hand side, we obtain $\sum_k \sum_\lambda (-1)^k (m_\lambda - b_\lambda) t^{\lambda+k} = \sum_\lambda q_\lambda t^\lambda$ which can be rewritten as $(\sum_k (-1)^k t^k)(\sum_\lambda (m_\lambda - b_\lambda) t^\lambda) = \sum_\lambda q_\lambda t^\lambda$. As an algebraic identity, we have that $\frac{1}{1+t} = \sum_k (-1)^k t^k$. Therefore, multiplying both sides by $1 + t$ we obtain the desired result.

$\square$

Appendices

Appendix A: Cellular Attachment Lemmas

In this appendix, we provide a proof of the cellular attachment lemmas concerning the extension of homotopy equivalences used in Section 2.4.

Lemma A.1. Let X be a topological space, $k \in \mathbb{N}$. Suppose $g_0, g_1 : \dot{e}^k \to X$ are homotopic attaching maps. Then, the identity map of X extends to a homotopy equivalence

$$h : X \cup_{g_0} e^k \to X \cup_{g_1} e^k.$$

Proof. Let g_t be the homotopy between g_0 and g_1, and let $X_0 := X \cup_{g_0} e^k$ and $X_1 := X \cup_{g_1} e^k$. Let $u \in \dot{e}^k$, which we interpet is $u \sim g_0(u)$ if $u \in X_0$ and $u \sim g_1(u)$ if $u \in X_1$. Let $t \in [0,1]$ such that $tu \in e^k$.

Let us consider the map $h : X_0 \to X_1$, which satisfies $h|_X = Id_X$ and in e^k is defined as

$$h(tu) = \begin{cases} 2tu & \text{for } 0 \leq t \leq \frac{1}{2}, \\ g_{2-2t}(u) & \text{for } \frac{1}{2} \leq t \leq 1. \end{cases}$$

This map h is continuous, since it is continuous on each part of its domain, and for $t = \frac{1}{2}$ we have $u \sim g_1(u)$.

A corresponding continuous map $l : X_1 \to X_0$ is defined by $l|_X = Id_X$ and

$$l(tu) = \begin{cases} 2tu & \text{for } 0 \leq t \leq \frac{1}{2}, \\ g_{2t-1}(u) & \text{for } \frac{1}{2} \leq t \leq 1. \end{cases}$$

Note that l is defined in the same way as h, but replacing $g_{\lambda(t)}$ with $g_{1-\lambda(t)}$ for the corresponding $\lambda(t)$ that appears in the defining expression.

We have that $l \circ h : X_0 \to X_0$ satisfies $l \circ h|_X = Id_X$ and

$$(l \circ h)(tu) = \begin{cases} 4tu & \text{for } 0 \leq t \leq \frac{1}{4}, \\ g_{4t-1}(u) & \text{for } \frac{1}{4} \leq t \leq \frac{1}{2}, \\ g_{2-2t}(u) & \text{for } \frac{1}{2} \leq t \leq 1. \end{cases}$$

Now, let $\xi_s : X_0 \to X_0$ with $s \in [0,1]$ be defined as $\xi_s|_X = Id_X$ and

$$\xi_s(tu) = \begin{cases} (4-3s)tu & \text{for } 0 \leq t \leq \frac{1}{4-3s}, \\ g_{(4-3s)t-1}(u) & \text{for } \frac{1}{4-3s} \leq t \leq \frac{2-s}{4-3s}, \\ g_{\frac{1}{2}(4-3s)(1-t)}(u) & \text{for } \frac{2-s}{4-3s} \leq t \leq 1. \end{cases}$$

Since $\xi_s(x)$ is continuous with respect to x and s, it defines a homotopy between $\xi_0 = l \circ h$ and $\xi_1 = Id_{X_0}$.

Similarly, one can compute $h \circ l : X_1 \to X_1$ and define a homotopy $\eta_s : X_1 \to X_1$ with the same expression that ξ_s but substituting $g_{\lambda(t)}$ with $g_{1-\lambda(t)}$ (as before). Then, we will have $\eta_0 = h \circ l$ and $\eta_1 = Id_{X_1}$.

□

Lemma A.2. Let X be a topological space, $k \in \mathbb{N}$. Let $\varphi : \dot{e}^k \to X$ be an attaching map. Then, any homotopy equivalence $f : X \to Y$ extends to a homotopy equivalence

$$F : X \cup_\varphi e^k \to Y \cup_{f\circ\varphi} e^k.$$

Proof. Let us define F by the conditions $F|_X = f$ and $F|_{e^k} = Id_{e^k}$. Let $g : Y \to X$ be a homotopy inverse to f and let us define $G : Y \cup_{f\circ\varphi} e^k \to X \cup_{g\circ f\circ\varphi} e^k$ by the corresponding conditions $G|_Y = g$ and $G|_{e^k} = Id_{e^k}$.

Since $g \circ f \circ \varphi$ and φ are homotopic attaching maps, by the previous Lemma 2.19 we have a homotopy equivalence

$$h : X \cup_{g\circ f\circ\varphi} e^k \to X \cup_\varphi e^k.$$

We want to prove that $h \circ G$ is a homotopy inverse of F, that is,

$$(h \circ G) \circ F \sim Id_{X\cup_\varphi e^k}, \tag{A.1}$$

$$F \circ (h \circ G) \sim Id_{Y\cup_{f\circ\varphi} e^k}. \tag{A.2}$$

First, let us prove that $h \circ G$ is a left homotopy inverse of F (relation (A.1)). Indeed, first note that if γ_t is a homotopy between $\gamma_0 = g \circ f$ and $\gamma_1 = Id_X$, by the definition of F, G and h (the one given in Lemma 2.19), we have $h \circ G \circ F|_X = g \circ f$ and for $u \in \dot{e}^k$

$$(h \circ G \circ F)(tu) = \begin{cases} 2tu & \text{for } 0 \leq t \leq \frac{1}{2}, \\ (\gamma_{2-2t} \circ \varphi)(u) & \text{for } \frac{1}{2} \leq t \leq 1. \end{cases}$$

Now, let us define the map $\beta_s : X \cup_\varphi e^k \to X \cup_\varphi e^k$ with $s \in [0,1]$ by $\beta_s|_X = \gamma_s$ and

$$\beta_s(tu) = \begin{cases} \frac{2}{1+s} tu & \text{for } 0 \leq t \leq \frac{1+s}{2}, \\ (\gamma_{2-2t+s} \circ \varphi)(u) & \text{for } \frac{1+s}{2} \leq t \leq 1. \end{cases}$$

Note that this map $\beta_s(x)$ is continuous with respect to x and s, and defines a homotopy between $\beta_0 = h \circ G \circ F$ and $\beta_1 = Id_{X \cup_\varphi e^k}$.

Now, we want to prove that $h \circ G$ is a right homotopy inverse of F (relation (A.2)). In this case, we do not explicitly construct a homotopy, but we use the following claim:

Claim: If a map F has a left homotopy inverse L and a right homotopy inverse R, then F is a homotopy equivalence and both R and L are homotopy inverses.

Indeed, this assertion is true since if $F \circ R \sim Id$ and $L \circ F \sim Id$, then we have:

$$L \sim L \circ (F \circ R) = (L \circ F) \circ R \sim R$$

which implies

$$R \circ F \sim L \circ F \sim Id, \quad F \circ L \sim F \circ R \sim Id.$$

We complete the proof in two final steps:

Step 1: By relation (A.1) we have that $G \circ F$ is a right homotopy inverse of h. Since h is a homotopy equivalence by hypothesis, it also has a left homotopy inverse and then by the previous claim $G \circ F$ is also a left homotopy inverse of h.

Step 2: We have seen that F has a left homotopy inverse and a similar proof shows that G has also a left homotopy inverse. Then, since in the previous step we have seen that $F \circ h$ is a right homotopy inverse of G, by the claim $F \circ h$ is also a left homotopy inverse of G. This proves the relation (A.2) and completes the proof.

□

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